Clique partitions with rooted simplicial defect: quantitative stability and sharp eventual bounds

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Abstract

We study how a large clique partition number constrains the structure of a graph and of its near-optimal partitions. For every fixed rooted simplicial defect s, we prove quantitative stability in the model c_4, where partition pieces have order at most four. If $c_4(G) \geq Q_s(n) - \delta$, with $0 \leq \delta \leq \gamma_s n^2$ and n above an explicit threshold, the graph is within $A_s\delta$ edge edits of a defect template on an actual clique root. That same root controls every near-optimal unrestricted clique partition, both in its number of noncanonical pieces and in the total number of edges contained in them. Approximation by a template of optimal size has a separate bound $A_s\delta + n\sqrt{(1+4A_s)\delta}$. An explicit family has distance at least $n\sqrt{\delta}/8$ from every optimal-size template, showing why the two forms of approximation must be distinguished.

For chordal graphs, the specialized constants are 16δ edits, $\tau + 48\delta$ noncanonical pieces, and $10\tau + 480\delta$ edges in those pieces when the partition has at most $M(n) + \tau$ parts. Here

$$M(n) = \left\lfloor \frac{n(n+1)}{6} \right\rfloor, \qquad Q_s(n) = M(n+s) - \binom{s+1}{2}.$$

Our construction attains the eventual upper bound $Q_s(n)$ using pieces of order at most four and recovers the unrestricted extremal family of Okechukwu [15, Theorem 1.1]. For chordal graphs it gives a proof of $c_4(G) \leq M(n) + b$ at every order, and hence of the bound asked in Erdős Problem 81. The all-orders additive bound for unrestricted clique partitions is also obtained in [5] and [15, Corollary 1.2]. No priority for the extremal value is claimed.

For sublinear rooted defect, we obtain an order-four approximation and stability along arbitrary sequences of orders tending to infinity, again with one root controlling all near-optimal partitions. The proof combines mixed rounding from the infrastructure of Papers I–III, retained constructor accounts, and explicitly attributed localization and removal arguments. Thresholds are explicit but impractical; neither optimality of the stability constants nor $b = 0$ for all chordal graphs is asserted.

Keywords: chordal graph; clique partition; mixed packing; fractional rounding; complete-split graph.

1. Introduction

A clique partition of a graph G is a family of cliques whose edge sets are disjoint and whose union is $E(G)$. We write $\mathrm{cp}(G)$ for the smallest number of pieces. Vertices may belong to several pieces; edges may not. For $r \geq 2$, we denote by $c_r(G)$ the minimum when every piece has order at most r.

The problem of Erdős, Ordman and Zalcstein [1,8] asks for a bound of the form $n^2/6 + O(n)$ for chordal graphs. Complete-split examples explain the quadratic coefficient. A fractional bound of this order, however, does not immediately yield a partition with linear error: general rounding allows a subquadratic loss, which may be much larger than n.

The three preceding papers in the series separate aspects of this difficulty. Paper I [2] studies neighborhood profiles relative to a split presentation and reduces the residual problem to a fixed polytope. Paper II [3] determines the exact maximum of the fractional triangle deficit over all chordal graphs by vertex copying. Paper III [4] constructs partitions with linear error for split graphs, distinguishing the regimes in which the fractional margin suffices from those requiring a more precise construction. Its formal development also supplies the nibble and rounding infrastructure reused in Section 3 and Appendix B.

The central question here is structural: what does near equality force in the graph, and how much of that structure must every near-optimal partition respect? The same clique root will answer both questions.

The main structural result retains information that a partition-count bound alone discards. In the chordal case, Theorem 6.1 gives the edit bound 16δ; the same root controls $\tau + 48\delta$ noncanonical pieces and at most $10\tau + 480\delta$ edges in them. Section 6.6 extends the same-root conclusion to every fixed rooted defect, with explicit constants and thresholds, and separates approximation by a defect template from approximation by a template of optimal size. Section 6.7 treats sublinear defect uniformly, rather than substituting a growing parameter into a fixed-defect theorem. The explicit threshold for mixed rounding also supplies the all-order additive constant in §6.3.

The public preprint [5] also establishes the eventual maximum for chordal graphs, which implies an all-orders additive bound. Okechukwu [15, Theorem 1.1] proves the eventual maximum for every fixed rooted simplicial defect, together with an all-orders additive bound for unrestricted clique partitions and an equality classification; its chordal case is [15, Corollary 1.2]. No priority is claimed for these statements. Theorem C proves the upper bound with pieces of order at most four, and Theorem C' recovers the equality classification. Together with Corollary 6.3, it gives a quantitative form of the qualitative edit stability in [15, Corollary 5.5]. Theorem 6.5 recovers the structural conclusion of [15, Theorem 1.3] and adds simultaneous control of near-optimal partitions.

Proposition 6.4, and consequently the approximation in Theorem 6.5, follows the signed-star argument of [15, Lemma 3.1]. The chordal-edit step follows de Joannis de Verclos [22]. Section 8 specifies the comparisons with [5,15] and the scope of these attributions.

Cipollini [21] announced the partial bound $n^2/6+o(n^2)$ for chordal graphs. That announcement identifies the remaining gap between subquadratic and linear error; it is an antecedent, not an input to our proof.

The formulation in [8] asks: "Can the edges of G be partitioned into $n^2/6+O(n)$ many cliques?", for chordal G. Erdős, Ordman and Zalcstein [1] had proved a bound $(1/4-\epsilon)n^2$, with a small positive constant ϵ. Theorem A gives the requested form for every order.

Here we extend the distinction between margin and construction to chordal graphs, using a

mixed model of triangles and K_4. The second type of piece has six edges and replaces six individual pieces by one. Its gain is therefore five, whereas a triangle's gain is two. These gains must be retained during rounding. Preserving only the number of copies does not preserve the cost of the partition.

Theorem A. Bound for all orders

There is an absolute constant $b \in \mathbb{N}$ such that, for every $n \geq 0$ and every chordal graph G of order n,

$$\operatorname{cp}(G) \leq c_4(G) \leq M(n) + b \leq \frac{n^2}{6} + \frac{n}{6} + b. \tag{A}$$

Consequently, there exists $C \geq 0$ such that $c_4(G) \leq n^2/6 + Cn$ for all these graphs. This gives the bound asked in Erdős Problem 81 [1,8] for every order, with no exceptions. The bound for unrestricted clique partitions is also obtained in [5] and [15, Corollary 1.2], as discussed in §8.1. The additive constant b is not claimed to be optimal.

Theorem B. Sharp bound and eventual maximum

There is an integer N such that every chordal graph G of order $n \geq N$ admits a partition into cliques of order at most four with at most $M(n)$ pieces. In particular,

$$\operatorname{cp}(G) \leq c_4(G) \leq M(n). \tag{1.1}$$

For the same orders, increasing N if necessary,

$$\max_{\substack{|V(G)|=n \\ G \text{ chordal}}} \operatorname{cp}(G) = M(n). \tag{1.2}$$

The second assertion follows from the upper bound and a complete-split witness whose optimality is proved against all clique partitions, not merely those of bounded order. This argument alone does not classify all extremal graphs. Theorem B is more precise than the bound requested in the problem; Theorem A states its consequence valid for every order.

The example that fixes the scale is explicit: for $n \geq 6$, take $k = \lceil n/3 \rceil$, $h = n-k$ and $S = K_k \vee I_h$. Eliminating the independent hosts first and then the core gives a perfect elimination ordering, so S is chordal. Using one core base per triangle yields a partition of cost $kh - \binom{k}{2} = M(n)$; the weight argument in §6.1 shows that no partition, even one with larger cliques, costs less.

The construction gives more than the extremal value. Theorem 6.1 bounds the edits needed to reach a complete-split graph, while Corollary 6.1a uses the same root, chosen independently of the partition, to control the pieces of every near-optimal partition. Setting the deficit to zero also identifies the eventual extremal graphs.

Theorem C. Fixed rooted simplicial defect

Following Okechukwu [15, §1], we say that G has **rooted simplicial defect at most** s if it satisfies the following condition. Given an induced vertex set $U \subseteq V(G)$ and a prescribed clique $R \subsetneq U$, there is a vertex $v \in U \setminus R$ whose neighborhood in U contains a clique omitting at most s of its neighbors. The empty root $R = \varnothing$ is allowed. Quantification over all prescribed roots is part of the definition, not a consequence of having a single elimination ordering. We also retain the notation Q_s from [15, (1.1)].

For each fixed integer $s \geq 0$, there is N_s such that every graph G in this class of order $n \geq N_s$ satisfies

$$\operatorname{cp}(G) \leq c_4(G) \leq Q_s(n), \qquad Q_s(n) := M(n+s) - \binom{s+1}{2}. \tag{C}$$

Moreover, for each sufficiently large order there is a graph in the class with $\mathrm{cp}(G) = Q_s(n)$. Thus the eventual maximum is $Q_s(n)$, also when cliques of arbitrary order are allowed.

This is an order-at-most-four strengthening of the eventual upper bound in [15, Theorem 1.1]. Proposition D.2 shows that four cannot be replaced by three uniformly over the classes in Theorem C: the obstruction already occurs for $s = 0$. It does not assert minimality separately for each positive defect.

The unrestricted maximum and equality family are those of [15, Theorem 1.1]; no priority is claimed for them. Every chordal graph belongs to the class $s = 0$, and $Q_0 = M$. Appendix E proves Theorem C through the explicit localization chain. Corollary 6.6 gives a tower bound for its threshold, uniform in the defect. The stability threshold below is specified separately; the exact bound and the stability bound must not be assigned the same threshold without a further comparison.

Theorem C'. Fixed-defect same-root stability

For each fixed $s \geq 0$, there are explicit N_s^{stab}, $\gamma_s > 0$, and $A_s = 2 \cdot 10^{41}(s+1)^8$ with the following property. Suppose $n \geq N_s^{\mathrm{stab}}$, $\mathrm{rsd}(G) \leq s$, and $c_4(G) \geq Q_s(n) - \delta$, where $0 \leq \delta \leq \gamma_s n^2$. Then $V(G)$ has a partition $C \sqcup D \sqcup H$, with C a clique of G, $|D| = s$, and $2 \leq |C| \leq |H|$, such that the template $T = (K_C \sqcup I_D) \vee I_H$ satisfies

$$d_E(G,T) \leq A_s\delta, \qquad 0 \leq Q_s(n) - \left(k(n-k) - \binom{k-s}{2}\right) \leq (1+4A_s)\delta, \quad k = |C| + s.$$

For this same root, every unrestricted clique partition with at most $Q_s(n)+\tau$ pieces has at most $\tau + (1+7A_s)\delta$ noncanonical pieces, containing at most $10\tau + 10(1+7A_s)\delta$ edges in total. Here the canonical pieces are the edges between $C \cup D$ and H, and the triangles with two vertices in C and one in H.

Section 6.6 gives the parameters and proof. At zero deficit this recovers the equality classification; at positive deficit the core need not have optimal size. Corollary 6.3 quantifies the additional cost of imposing optimal size. Proposition 6.3a supplies examples at the square-root scale, including graphs of exact deficit one whose distance from every optimal-size template grows with the order. These fixed-defect constants are not substituted into a growing-s limit: §6.7 treats that limit separately.

1.1. The common condition: a loss budget

Let $W^*(G)$ be the optimal fractional mixed gain from triangles and K_4, with gains two and five. Write

$$F_4(G) = e(G) - W^*(G), \qquad \Delta(G) = M(n) - F_4(G).$$

For a physical packing $\mathcal{P}$ of gain $g(\mathcal{P})$, completing the uncovered edges as individual pieces costs exactly $e(G) - g(\mathcal{P})$. Hence

$$\boxed{e(G) - g(\mathcal{P}) \leq M(n) \quad \Longleftrightarrow \quad W^*(G) - g(\mathcal{P}) \leq \Delta(G)}. \tag{1.3}$$

The inequality on the right has a direct interpretation. The quantity $W^*(G) - g(\mathcal{P})$ measures the gain lost in passing from the fractional optimum to the constructed packing; $\Delta(G)$ measures the margin left by that optimum relative to $M(n)$. If the loss does not exceed the margin, the completed partition has at most $M(n)$ pieces. We call this comparison the **loss budget**. The identity does not specify how to choose $\mathcal{P}$: it can be used to check any construction of edge-disjoint copies.

The accounting has a form independent of the mixed model. Let Q be any clique partition, with pieces of order at least two, and define its total gain by

$$g(Q) = \sum_{K \in Q} \left(\binom{|K|}{2} - 1 \right).$$

Exact coverage and edge-disjointness give, for any target T,

$$g(Q) + |Q| = e(G), \qquad |Q| \le T \quad \Longleftrightarrow \quad e(G) \le g(Q) + T. \tag{1.3a}$$

No bound on the order of the pieces has been used. In particular, K_2 pieces have zero gain and may be retained in the sum or added when completing a packing. For large pieces one uses the ordinary gain, not the capped function of the mixed model.

To compare families, fix a set $S \subseteq \{3, \dots, n\}$ of allowed orders. Let $W_S^*(G)$ denote the maximum fractional gain with these pieces, $F_S(G) = e(G) - W_S^*(G)$ the fractional cost, and $\Delta_S(G) = M(n) - F_S(G)$ its margin. Capacities remain one per edge. Extending a packing by zero shows that, if $S \subseteq S'$, then $W_S^* \le W_{S'}^*$ and $\Delta_S \le \Delta_{S'}$. For $4 \le L \le n$, this gives

$$\Delta_{\{3,4\}}(G) \le \Delta_{\{3,\dots,L\}}(G) \le \Delta_{\mathrm{cp}}(G), \tag{1.3b}$$

where the last term permits every order up to n. With the same convention, $c_4(G) \ge c_L(G) \ge \mathrm{cp}(G)$. The margin inequalities need not be strict. Moreover, for a fixed construction, enlarging S increases both the optimum against which the loss is measured and the margin: the advantage is the availability of more admissible constructions, not an automatic improvement in the balance.

Thus (1.3) is the mixed case of a general accounting identity. The difficulty remains to produce a construction satisfying it for every chordal graph. The following structural result provides this coverage for our family of pieces.

We prove that uniform rounding RC01 produces a loss smaller than the margin in the far regime. In the critical regime, descent locates a root and the H1/RD09 construction produces $\mathcal{P}$ directly in G, without selecting a first-entry index. Its physical accounting, developed in Section 5, proves

$$e(G) - g(\mathcal{P}) \le B_n(p) - \frac{m}{20} - \frac{A}{2} \le M(n), \tag{1.4}$$

and (1.3) gives the required budget.

We fix three terms. The **margin** is $\Delta(G)$; the **loss** is $W^*(G) - g(\mathcal{P})$. **Selector slack**, by contrast, refers to unused capacity in an auxiliary fractional packing. It does not require a lower bound on loads and should not be confused with the graph's margin.

1.2. Structural dichotomy and resolution mechanisms

Structural dichotomy. For $\eta_0 = 10^{-16}$ and every sufficiently large chordal graph,

$$\boxed{F_4(G) < \frac{n^2}{6} - \eta_0 n^2 \quad \text{or} \quad \exists\, \text{near-structure witness for } G}. \tag{1.5}$$

The witness includes a complete-split comparator, a root that is a clique of G, and a packing of the original graph with its physical accounting. The alternatives are exhaustive; the existence of the witness is not asserted to be incompatible with a quadratic margin. This is not merely a metric condition: it retains the resources needed to complete the partition.

In the first branch, mixed rounding produces a loss smaller than the available margin in (1.3). In the second, the constructor's accounting verifies the target directly. The packing, its feasibility

and the accounting inequality suffice for the conclusion. The near witness also retains the structure that produces them.

Figure 1 follows only our proof. To connect the diagram to the text, we call the far case **branch L**: Theorem 3.1 performs rounding and Corollary 3.5 pays for its loss. **Branch C** is the critical case: Proposition 4.3 locates the comparator, Section 5 constructs in G, and Lemma 5.3 pays for the losses. Both branches end in Theorem B; Section 6.3 extends its conclusion to every order and proves Theorem A. The letters L and C denote the two cases of a single proof.

The root and the accounts in the critical branch are retained after the partition has been constructed. They lead to the graph and partition stability results of Section 6.5: the root is chosen from the graph and serves all near-optimal partitions simultaneously.

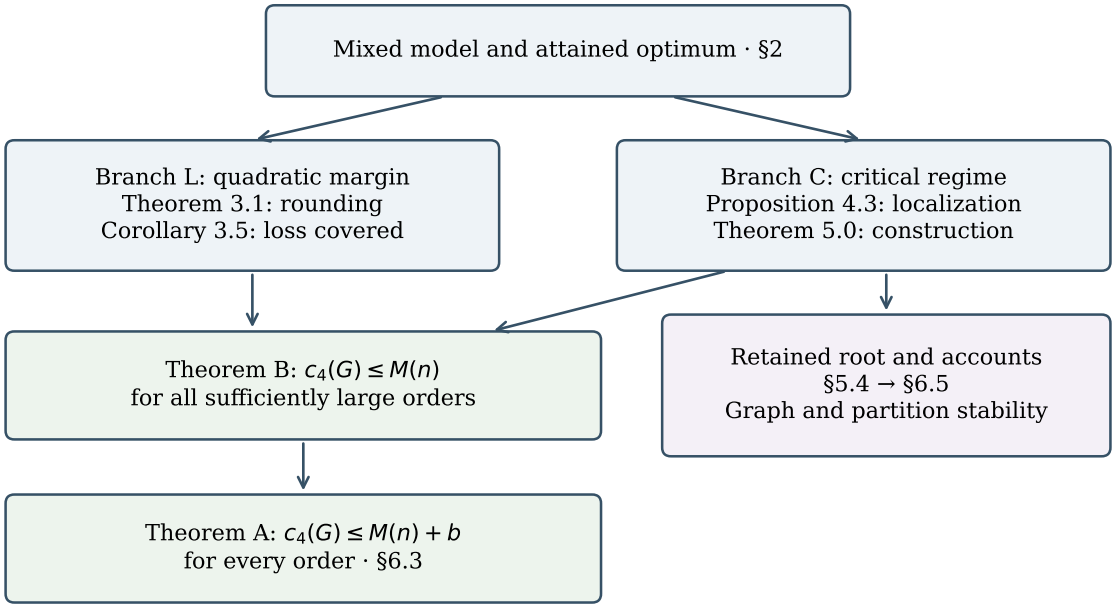


Figure 1: Proof map. The two branches construct a partition of the original graph. Retaining the root and its accounts in the critical branch also gives the stability results.

1.3. Organization and use of chordality

Remark 1.1. The rounding branch applies to arbitrary graphs. The critical branch uses chordality at three distinct points: the existence and preservation of the copy steps in §4.1; clique extraction and the absence of induced squares in §5.1; and the perfect elimination ordering that orients exterior edges in §5.3. The elimination ordering also certifies that the examples attaining the bound are chordal. None of these properties follows solely from the numerical budget inequalities.

Section 2 fixes the model. Section 3 states the rounding result and explains its use; Appendix C develops its technical proof. Sections 4 and 5 establish localization and the critical construction. Section 6 assembles the chordal proof and derives stability, classification and exact values. Sections 6.6 and 6.7 then extend stability to fixed and sublinear rooted defect; Appendix F records their parameters, proof interfaces and additional provenance. Section 7 points to the separate formal companion; Appendix A records the scope of the conclusions. Section 8 compares the mechanisms with [5,15]. Appendix B collects complementary tools. Appendix D also records the comparator identities, and Appendix G treats the bounded-clique triangle gap and the separate limitation of the cleanup contract.

The general transfer framework is developed in Paper V. It is not an additional input to the proof given here.

2. Mixed gain, duality and partitions

All graphs are finite, simple and undirected. A clique is identified with its vertex set; its edges are all pairs of distinct vertices in that set. A graph is chordal if it has no induced cycle of length at least four. Induced subgraphs of a chordal graph are chordal.

2.1. Why gain is the quantity to round

Let $\mathcal{K}(G)$ be the family of cliques of order three or four. A mixed packing is a subfamily $\mathcal{P} \subseteq \mathcal{K}(G)$ of edge-disjoint copies. Define

$$g(K) = \binom{|K|}{2} - 1, \qquad g(\mathcal{P}) = \sum_{K \in \mathcal{P}} g(K).$$

Piece	Edges covered	Pieces replaced	Gain
K_2	1	1	0
K_3	3	3	2
K_4	6	6	5

Table 1. Cost is measured relative to covering each edge separately.

Lemma 2.1

Let Q be the partition obtained by adding to $\mathcal{P}$ every uncovered edge as a K_2 piece. Its number of pieces is exactly

$$|Q| = e(G) - g(\mathcal{P}). \tag{2.1}$$

Proof. The pieces of $\mathcal{P}$ cover $\sum_K \binom{|K|}{2}$ edges, since they do not overlap. There remain $e(G) - \sum_K \binom{|K|}{2}$ individual edges. Adding the initial $|\mathcal{P}|$ pieces gives (2.1). Coverage is exact by construction.

The mixed gain is two for a triangle and five for a clique of order four; extending that function by zero to other orders is only a convention for the mixed model. An unrestricted partition instead uses $\binom{|K|}{2} - 1$ for every piece. The two conventions agree on the mixed pieces but differ from order five onwards. Since every partition piece has at least two vertices, subtraction in this expression is ordinary subtraction. This distinction is needed when applying the accounting identity to unrestricted partitions in §6.4.

2.2. The fractional optimum and its certificate

A fractional packing assigns a weight $x_K \geq 0$ to each actual copy K, with

$$\sum_{K : e \in E(K)} x_K \leq 1 \qquad (e \in E(G)). \tag{2.2}$$

Its value is $w(x) = \sum_K g(K) x_K$, and $W^*(G)$ is the maximum of these values. The dual assigns prices $y_e \geq 0$ to edges. For each triangle T and each K_4, denoted by Q, it requires, respectively,

$$\sum_{e \in E(T)} y_e \geq 2, \qquad \sum_{e \in E(Q)} y_e \geq 5. \tag{2.3}$$

The zero vector is primal feasible. Every primal coordinate is bounded by one, since every copy contains an edge. Finite duality supplies primal and dual optima of equal value. The model uses rational data and constructs both optima: the value w is not an assumption without a witness. Paper I [2, Appendix B] provides the antecedent of packing–covering duality; the rational development used here proves its version by Fourier–Motzkin elimination.

Summing capacities in (2.2) also gives

$$w(x) \le \frac{5}{6} \sum_{K} |E(K)| x_K \le \frac{5}{6} e(G). \tag{2.4}$$

This estimate controls the cost of scaling a packing down multiplicatively. It is not, by itself, the chordal extremal bound.

2.3. Relation to the earlier functionals

Extending a triangular packing by zero gives

$$W^*(G) \ge 2\nu_3^*(G), \qquad F_4(G) \le e(G) - 2\nu_3^*(G). \tag{2.5}$$

Paper II [3, Theorem 1.1] bounds the second quantity by $\lfloor (2n+1)^2/24 \rfloor$. For integer n this floor equals $M(n)$: the product $n(n+1)$ is congruent to 0 or 2 modulo 6, and adding $1/24$ does not cross an integer. This explains the target of the series. Passing from that fractional value to a partition still requires the two branches developed below.

2.4. Notation for the two branches

Symbol	Meaning
$M(n)$, $B_n(p)$	Integer target and complete-split profile
$W^*(G)$, $W(G)$	Optimal fractional and integral mixed gains
$g(\mathcal{P})$	Gain of the chosen physical packing
$F_4 = e - W^*$, $\Delta = M - F_4$	Fractional cost and margin to the target
W_S^*, F_S, Δ_S	The same quantities for a family of orders S, only in §§1.1 and 8
C, P, R	Comparator core, extracted clique and regularized root
a, p, q	$\lvert P\rvert, \lvert R\rvert, n - \lvert R\rvert$
m, A	Exterior edges and missing links relative to R
f, ℓ	First-phase triangles and failed second-phase bases
$d(G)$	Normalized edit distance to the complete-split family

The auxiliary symbols for the selector in Appendix C are defined there. In particular, ℓ in §5 is a failure count, not a gain.

3. Mixed rounding and the regime with margin

This is branch L in Figure 1. Its input is a fractional packing; its output is a physical packing with uniformly subquadratic loss. Corollary 3.5 explains when this loss fits within the available margin.

Theorem 3.1

For every rational $\xi > 0$, there exists N_ξ such that, for every graph G of order $n \geq N_\xi$ and every rational fractional mixed packing x, there is a mixed packing $\mathcal{P}$ with

$$w(x) - g(\mathcal{P}) \leq \xi n^2. \tag{3.1}$$

Chordality is not required. The threshold is chosen after ξ, but before G, n and x.

This is a uniform transfer of weighted gain. The theorems of Haxell–Rödl and Yuster [6,7] provide the context of fractional-to-integral transfer. We specify below the input from Paper III and the two-quota adapter that preserves gains two and five.

Proof structure. The bounded-rank lemma of Paper III, reproduced as Lemma 3.2 in Appendix C, rounds a family with loads at most one and sufficiently small codegree. By itself, it does not distinguish gains two and five. Lemma 3.3 adds marks and obtains two quotas for the same matching; Lemma 3.4 groups pairs of triangles and marks the K_4 copies, so that the gain is recovered as four times the total quota plus the marked quota.

To apply that selector to an arbitrary graph, a regular partition and profile cleanup produce a system with controlled loads and codegrees. Removed mass is charged to its resources only once. First ξ is fixed, then the selector and regularity parameters, and only at the end the threshold N_ξ, independently of the instance. The case of small triangular mass is treated separately. Appendix C retains the full estimates and parameter hierarchy; its numbering 3.2–3.14 belongs to the proof of this theorem.

Thus Theorem 3.1 is not cited as a black box from Paper III. That paper supplies the nibble; two-quota selection, its mixed realization and the uniform assembly are the steps of Paper IV. The fully proved statement (3.1) is all that is used below.

Corollary 3.5

For a fixed rational $\eta > 0$, every sufficiently large graph with $F_4(G) < n^2/6 - \eta n^2$ admits a partition of order at most four with at most $M(n)$ pieces.

Proof. Apply Theorem 3.1 to a rational optimum with $\xi = \eta/2$. Lemma 2.1 gives

$$c_4(G) \leq e(G) - g(\mathcal{P}) \leq F_4(G) + \frac{\eta}{2} n^2 < \frac{n^2}{6} - \frac{\eta}{2} n^2.$$

For $n \geq 6$, the inequality $\lfloor t \rfloor \geq t - 1$ gives $M(n) \geq n(n+1)/6 - 1 \geq n^2/6$. The conclusion follows by increasing the threshold if necessary.

The argument does not require a universal linear bound on $W^*(G) - W(G)$. It only requires the loss to be smaller than the margin of the instance.

4. Localization of the critical regime

We begin branch C of Figure 1. Proposition 4.3 locates a complete-split comparator. Section 5 converts this information into a partition of the original graph.

Fix

$$\varepsilon_0 = 10^{-12}, \qquad \eta_0 = 10^{-16}. \tag{4.1}$$

The order threshold $4 \cdot 10^{12}$ used below applies to localization and the near construction. The rounding in Section 3 may require a larger order; this is not a global threshold.

4.1. Distance to a fixed family

For graphs on the same vertex set, let $d_E(G,H) = |E(G) \triangle E(H)|$. Consider the family of all labelled complete-split graphs $S_C = K_C \vee I_{V\setminus C}$, and define

$$d(G) = \frac{1}{n^2} \min_C d_E(G, S_C).$$

The minimum is attained because the family is finite. The triangle inequality implies

$$|d(G) - d(H)| \le \frac{d_E(G,H)}{n^2}. \tag{4.2}$$

A vertex-copy step changes at most $n-1$ edges. Thus the change in d in one step is at most $1/n$. This bound does not require the distance to decrease along the whole path.

Paper II [3] explains the mechanism of copying simplicial classes and termination at a complete-split graph for the triangular functional. The mixed development requires its own value transports and ends at a graph in the same family. Admissible copies preserve chordality; the mixed bound does not follow simply by renaming the triangular functional.

Lemma 4.1 (admissible mixed path). Every chordal graph G admits a finite sequence $G = G_0, \dots, G_\ell$ on the same vertex set such that G_ℓ is complete-split, every G_i is chordal, each step copies a single nonadjacent simplicial vertex, and

$$F_4(G_i) \le F_4(G_{i+1}), \qquad d_E(G_i, G_{i+1}) \le n-1. \tag{4.3}$$

Proof. For two nonadjacent simplicial vertices u, v, consider the two opposite copies $G_{u\to v}$ and $G_{v\to u}$. Mixed packing transport gives

$$W^*(G_{u\to v}) + W^*(G_{v\to u}) \le 2W^*(G). \tag{4.4}$$

This is the mixed version of the transport in Paper II. Both packings are transported to the original graph and averaged with weight $1/2$. A link may receive load from pieces transported from both endpoints; the preliminary bound is load at most two, not one. Averaging restores capacity one and also divides the sum of gains by two. The sum of the edge counts of the two copies is $2e(G)$. Subtracting (4.4) gives $2F_4(G) \le F_4(G_{u\to v}) + F_4(G_{v\to u})$; at least one direction does not decrease F_4.

It remains to show that a step always exists outside the terminal family and that steps do not repeat. Call two vertices twins if they have the same open neighborhood. If every nonadjacent simplicial pair were twins, Dirac's theorem would supply, in the noncomplete case, a pair x, y with a common neighborhood C, which is a clique. The exterior of C is independent: if it had a component containing an edge, the rooted form of Dirac's theorem would give a simplicial vertex of the graph in that component. Its neighborhood would have to be C, contradicting the presence of a neighbor in the component. Every exterior vertex is therefore simplicial and, by the condition on pairs, has neighborhood exactly C. Hence the graph is $K_C \vee I_{V\setminus C}$. The complete case is already terminal. Taking the contrapositive gives the two distinct classes required for a step.

Now count ordered pairs of distinct twins. Let $t \le s$ be the sizes of the two classes, and copy a vertex from the smaller class toward the larger one. Among vertices not copied, no equality of neighborhoods is destroyed: all undergo the same substitution in the coordinate of the copied vertex. At most $2(t-1)$ pairs containing it are lost and $2s$ are gained, so the increase is at least $2(s-t+1) > 0$. If this direction does not decrease F_4, it lexicographically increases the pair consisting of F_4 and this count. If it decreases F_4, (4.4) forces the reverse direction to increase

it strictly; that too increases the pair. There are only finitely many labelled graphs on V, so the process terminates.

Finally, deleting the vertex to be copied leaves a chordal induced subgraph. Reinserting it with the source's clique neighborhood adds a simplicial vertex; any cycle through it of length at least four has the chord between its two neighbors. Chordality is therefore preserved. Only edges incident to the copied vertex change, at most $n-1$. This proves (4.3).

Lemma 4.2 (accounting inside the window). If X is chordal, $n \geq 4 \cdot 10^{12}$, $F_4(X) \geq n^2/6 - \eta_0 n^2$ and $d(X) < \varepsilon_0$, then

$$d(X) \leq 16D/n^2, \qquad D = \eta_0 n^2 + n/6 + 1/24. \tag{4.5}$$

Proof. Apply the local construction of Theorem 5.0, whose proof does not use Proposition 4.3. Its strengthened account (5.15a) gives a clique root R and a partition Q with

$$F_4(X) \leq |Q| \leq B_n(|R|) - \frac{117}{1825}m - \frac{12687}{20000}A.$$

The first inequality holds because the fractional optimum dominates the gain of the physical packing. By (5.16), $B_n(|R|) \leq (2n+1)^2/24$. Subtracting the assumed lower bound on $F_4(X)$ gives

$$\frac{117}{1825}m + \frac{12687}{20000}A \leq D.$$

Since $16(117/1825) > 1$ and $16(12687/20000) > 10$, we obtain $m + 10A \leq 16D$. As R is a clique, $d_E(X, S_R) = m + A \leq 16D$, proving (4.5). The strengthened account belongs to the **local construction under proximity**; global localization is proved next using this lemma. Thus there is no circular dependency.

4.2. Descent from the terminal graph

The localization argument can be viewed as backward induction. Suppose $G_0, \ldots, G_\ell$ is an admissible path and G_ℓ is complete-split. Its distance is zero. The local accounting estimate ensures that, within the window $d(X) < \varepsilon_0$, graphs on the path with the required fractional value satisfy

$$d(X) \leq \frac{16D}{n^2} < \varepsilon_0 - \frac{1}{n}, \qquad D = \eta_0 n^2 + \frac{n}{6} + \frac{1}{24}. \tag{4.6}$$

The calibration verifies the last inequality for $n \geq 4 \cdot 10^{12}$.

Specifically, the inequality to check is

$$\frac{16}{6n} + \frac{1}{n} + \frac{16}{24n^2} < \varepsilon_0 - 16\eta_0. \tag{4.6a}$$

The change has two concrete causes. The account (5.15a) reduces the budget factor from 20 to 16; the descent also permits the maximal inner barrier $\varepsilon_0 - 1/n$, without separately reserving the fractions $\varepsilon_0/2$ and $\varepsilon_0/4$. Inequality (4.6a) holds at $n = 4 \cdot 10^{12}$, and its left-hand side decreases with n. The contraction inequality with this barrier fails for $n \leq 3.6 \cdot 10^{12}$ at the current values of ε_0, η_0 and the factor 16. This limit concerns this calibration, not every possible method.

If $d(G_{i+1}) < \varepsilon_0 - 1/n$, then (4.2) gives

$$d(G_i) < \varepsilon_0.$$

We may now apply (4.6) to G_i, returning it to the interval $d(G_i) < \varepsilon_0 - 1/n$. Repeat this step back to the original graph. The argument uses an outer window and an inner contraction; it does not require selecting the first index that crosses a threshold.

This induction does not eliminate the local stability estimate (4.6). It identifies precisely the input that replaces first-entry selection by descent. Monotonicity of the fractional value preserves the numerical hypotheses even when the distance oscillates.

The same propagation can be proved by deleting the last step and inducting on the path length. These are two proofs of the localization lemma, not two additional global solutions. Neither requires selecting a minimum index.

The near threshold reflects the interaction of two scales: removing u vertices from the comparator costs at most un, whereas chordal accounting controls $u(u+1)/2$ by the number of edits. A parameterized version takes a budget a^2/B with $B \geq 1$, a bound $a \geq \alpha n$ with $0 < \alpha \leq 1$, and $\eta \geq 0$. The conditions

$$\varepsilon = \frac{\alpha^4}{8B^2}, \qquad 1280B^2\eta \leq \alpha^4, \qquad n \geq \frac{1280B^2}{\alpha^4}$$

are sufficient for the numerical contraction in the descent. This scale is not claimed to be necessary or optimal. The specific calibration used here exploits sharper estimates and gives the near threshold stated in Proposition 4.3.

Proposition 4.3

Let G be chordal of order $n \geq 4 \cdot 10^{12}$, and suppose

$$F_4(G) \geq n^2/6 - \eta_0 n^2.$$

There is a nonempty set $C \subseteq V(G)$ such that

$$d_E(G, S_C) \leq \varepsilon_0 n^2, \tag{4.7}$$

$$(6|C| - 2n - 1)^2 \leq 24\left((\eta_0 + 6\varepsilon_0)n^2 + \frac{n}{6} + \frac{1}{24}\right). \tag{4.8}$$

The preceding localization and the comparator estimate give (4.7)–(4.8). The set C is not asserted to be a clique of G: it is a clique of the comparator S_C. The next construction obtains a root that is a clique of the original graph. This distinction avoids transporting a partition through an edit set that may have quadratic size.

5. Regularized root and physical accounting

Theorem 5.0. Near constructor under local hypotheses

Let $\varepsilon_0 = 10^{-12}$, $\eta_0 = 10^{-16}$, and let G be chordal of order $n \geq 4 \cdot 10^{12}$. Suppose

$$d(G) < \varepsilon_0, \qquad F_4(G) \geq n^2/6 - \eta_0 n^2.$$

There are a clique R of G of size p with $|3p - n| \leq n/50$, and a partition Q of order at most four such that, if $m = e(G - R)$ and A counts missing links between R and its exterior,

$$|Q| \leq B_n(p) - \frac{m}{20} - \frac{A}{2}, \qquad B_n(p) = p(n-p) - \binom{p}{2}. \tag{5.0}$$

Proof and organization. Lemma 5.1 extracts an actual clique; Proposition 5.2 regularizes it. The two phases in §§5.2–5.3 construct a packing on the same resources whose completion satisfies Table 2. Lemma 5.3 gives (5.0). For the size window, Lemma 5.1 gives $a \geq 33n/100$. The regularization accounts give $99a \leq 100p$ and $48(2p - (n-p)) \leq a$, with the subtraction

truncated before passing to the rational inequality. Together with $p + (n - p) = n$, these imply $3267n \le 10000p$ and $3539p \le 1188n$, hence $|3p - n| \le n/50$. In particular, the earlier window $n/4 \le p \le n/2$ remains valid. None of these steps uses the global localization of Proposition 4.3. The present theorem can therefore be applied in Lemma 4.2, and localization obtained only afterwards. The following subsections prove each part of the statement.

5.1. Regularization on the original graph

The construction in this section is local: it takes a chordal graph G already satisfying $d(G) < \varepsilon_0$ and $F_4(G) \ge n^2/6 - \eta_0 n^2$. It can therefore be used in Lemma 4.2 before global localization is established. Once Proposition 4.3 is proved, it applies to the original graph.

Lemma 5.1 (extraction of an actual clique). Under these hypotheses and $n \ge 4 \cdot 10^{12}$, there is a clique P of G of size a such that

$$\begin{gathered} a \ge 1024, \qquad |n - 3a| \le a/64, \\ e(G - P) + \#\{\text{missing links between } P \text{ and } V \setminus P\} \le a^2/65536. \end{gathered} \tag{5.1}$$

Proof. Let C be a core of the nearest comparator and write $r = |C|$. Fractional robustness under edits gives $|F_4(G) - F_4(S_C)| \le 6 d_E(G, S_C)$. Set $\delta_C = (\eta_0 + 6\varepsilon_0)n^2 + n/6 + 1/24$; the hypothesis implies $F_4(S_C) \ge (2n+1)^2/24 - \delta_C$.

We explain what happens outside the split regime with many hosts. Write $a_0 = \binom{r}{2}$, $b_0 = r(n-r)$. Explicit fractional packings of the comparator give the bounds

$$F_4(S_C) \le \begin{cases} b_0 - a_0, & b_0 \ge 2a_0, \\ (2b_0 - a_0)/3, & a_0 \le b_0 \le 2a_0, \\ (a_0 + b_0)/6, & b_0 \le a_0. \end{cases} \tag{5.1a}$$

These three constructions apply for $r \ge 4$ and a nonempty exterior. The middle branch is bounded by $(4n+1)^2/120$, and the third by $n(n-1)/12$. For $n \ge 9$, both lie strictly below $(2n+1)^2/24 - n^2/40$. Since here $\delta_C \le n^2/40$, neither satisfies the fractional proximity hypothesis. If $r < 4$, the bound $F_4(S_C) \le 3n$ suffices; if the exterior is empty, the uniform K_4 packing gives $F_4(S_C) \le n(n-1)/12$. For $n \ge 100$, these cases are also excluded. The first branch remains: $F_4(S_C) \le b_0 - a_0 = B_n(r)$. Completing the square then gives $(6r - 2n - 1)^2 \le 24\delta_C$. This justifies the estimate without extending a split identity to core sizes where it does not apply.

Now take a maximum clique P of the chordal induced subgraph $G[C]$, and put $u = r - a$. A perfect elimination ordering bounds its edges by $(a-1)r - \binom{a}{2}$. Consequently, at least

$$\binom{r}{2} - \left((a-1)r - \binom{a}{2}\right) = \binom{u+1}{2}$$

pairs are missing inside C. All are counted among the comparator edits, so $u(u+1)/2 \le \varepsilon_0 n^2$. In particular, $u \le 3n/(2 \cdot 10^6)$. Removing these u vertices from the comparator core changes at most un additional pairs. Hence the exterior and cross defects relative to P sum to at most $\varepsilon_0 n^2 + un$.

To see the scale of the calibration, $\delta_C \le (61/10)\varepsilon_0 n^2$ places r at distance less than $3 \cdot 10^{-6} n$ from $n/3$. Together with the bound on u, this implies $a \ge 33n/100$ and $|n - 3a| \le a/64$. Moreover,

$$\varepsilon_0 n^2 + un \le (10^{-12} + 1.5 \cdot 10^{-6})n^2 \le \frac{(33n/100)^2}{65536} \le \frac{a^2}{65536}.$$

These are rational comparisons with the constants fixed in (4.1). The threshold on n also gives $a \ge 1024$. This proves (5.1) without ever treating C as a clique of G.

The final root is defined explicitly. Remove from P the vertices whose column of missing links has size at least $|P|/4$, and add exterior vertices of exterior degree at least $7|P|/4$. If these sets are X and Y, respectively, then

$$R = (P \setminus X) \cup Y. \tag{5.2}$$

The defect budget gives

$$16384|X| \le |P|, \qquad 57344|Y| \le |P|. \tag{5.3}$$

Proposition 5.2 (regularization). The root R in (5.2) is a clique. If $p = |R|$, $q = n - p$, $m = e(G - R)$, A is its cross defect and D is the maximum defect of a column, it satisfies the palette and width bounds in (5.7), and

$$\begin{gathered} 99a/100 \le p \le 101a/100, \quad q \ge p, \quad 44352q \ge 87947a, \\ 48(2p - q)_+ \le a, \quad 3D \le a, \quad 400m < a^2, \quad 2000(A + 2m) \le 11a^2. \end{gathered} \tag{5.4}$$

Proof. The sum of the missing columns is part of budget (5.1), so $(a/4)|X| \le a^2/65536$. The sum of exterior degrees is twice the number of exterior edges and gives $(7a/4)|Y| \le 2a^2/65536$. These are the two bounds in (5.3).

Every clique exterior to P has at most $a/128 + 1$ vertices: its internal pairs count toward the same budget. In a chordal graph, common neighbors of two nonadjacent vertices form a clique, since two nonadjacent common neighbors would complete an induced square. If two members of Y were nonadjacent, their exterior neighborhoods would intersect in at least $7a/2 - 129a/64 = 95a/64$ vertices, contradicting the exterior clique bound. Here we used $|V \setminus P| \le 129a/64$, which follows from (5.1). Thus Y is a clique. If $x \in P$ is nonadjacent to a member of Y, the same property bounds their common exterior neighbors by $a/128 + 1$. The exterior degree of the member of Y then forces at least $a/4$ missing links in the column of x; hence $x \in X$. This proves that R is a clique.

For the remaining bounds, count the resources changed by removing X and adding Y. If m_P, A_P are the defects relative to P, then $m \le m_P + |X|n$ and $A \le A_P + |Y|n$. Exterior degrees and width satisfy

$$64d_{\max}(G - R) \le 113a + 64|X|, \qquad 128\omega(G - R) \le a + 128 + 128|X|.$$

The coefficient $113/64$ comes from the removed columns: a vertex of X misses at least $a/4$ of the at most $129a/64$ old exterior vertices, and thus has at most $113a/64$ neighbors there. A vertex of the old exterior outside Y has degree below $7a/4 \le 113a/64$ there. Moving X adds at most $|X|$ neighbors in either case. The retained columns start with fewer than $a/4$ missing links; added columns are controlled by the same common-neighbor argument. Substituting (5.3), $a \ge 1024$ and $127a \le 64|V \setminus P| \le 129a$ into these counts gives (5.4) and (5.7). These are the bounds used below: the defect is not chosen anew after the packing has been constructed.

The purpose of (5.2) is not to minimize an abstract energy. It separates the two defects that would prevent host assignment: columns with too many missing links and exterior vertices with too much residual degree. The result retains explicit bounds on size, exterior width and link availability.

5.2. From matchings to triangles

A matching of bases uv, together with a vertex z adjacent to all their endpoints, produces the triangles zuv (Figure 2). They are edge-disjoint: no endpoint repeats within the matching, so no link to z repeats either.

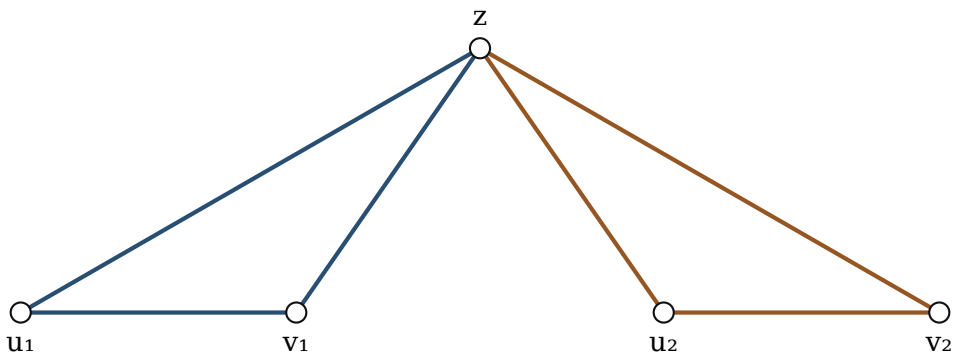


Figure 2: Two disjoint bases with a common host produce two triangles sharing a vertex but no edge. This is the physical unit used in assignments by color class.

With several hosts, we additionally require them to be distinct, no host to be an endpoint of a base, and the base families to be disjoint. These conditions prove compatibility between classes; gains are not added for constructions whose disjointness has not been verified.

The first phase applies this realization to exterior edges, using root vertices as hosts. A balanced coloring and the rooted elimination ordering control how many bases can be hosted. The second phase factors the root edges into matchings and assigns exterior candidates to them. Candidate lists exclude both missing links and resources already used by the first phase.

5.3. The three accounts that pay for the construction

To compare the actual construction with the complete-split model, first fix the root size. Put $p = |R|$ and $q = n - p$. If all edges between the root and its exterior were present and there were no exterior edges, the reference cost would be

$$B_n(p) = p(n-p) - \binom{p}{2}.$$

Indeed, covering the complete-split edges individually would cost $p(n-p) + \binom{p}{2}$ pieces. Each of the $\binom{p}{2}$ internal bases is included in a triangle with an exterior host; that triangle replaces three individual edges by one piece, saving two. The resulting cost is therefore $p(n-p) + \binom{p}{2} - 2\binom{p}{2} = B_n(p)$. With this reference fixed **before** constructing the packing, introduce the actual deviations. Let m be the number of edges exterior to R, A the number of missing links between R and its exterior, f the number of first-phase triangles, and ℓ the number of failed second-phase bases. If Q is the physical completion, the construction lemmas give the following three relations.

Account	Identity or inequality	Role
Piece count	$\lvert Q\rvert + A + 2f = B_n(p) + m + 2\ell$	Expresses the actual cost
First phase	$1600m \le 2920f + 219A$	Pays for exterior edges
Second phase	$200\ell \le 35A + 8f$	Bounds failed bases

Table 2. All loss quantities come from finite families. The reference value $B_n(p)$ is fixed before counting Q.

The first-row identity follows by counting. There are $e(G) = \binom{p}{2} + pq + m - A$ edges. The first phase contributes f triangles and the second $\binom{p}{2} - \ell$. Since the families are compatible, Lemma

2.1 gives

$$|Q| = \binom{p}{2} + pq + m - A - 2\left(f + \binom{p}{2} - \ell\right) = B_n(p) + m - A - 2f + 2\ell. \tag{5.4a}$$

Rearranging gives the first account. In particular, we do not separately estimate the costs of two constructions that might compete for the same edges.

Compatibility is checked explicitly. If a first-phase triangle uses an exterior edge uv and a host $z \in R$, it consumes the links zu, zv. The second phase cannot assign u or v to a base containing z. Together with adjacency to both endpoints of the base, this exclusion ensures that surviving assignments do not reuse links consumed in the first phase. The two triangle families are therefore edge-disjoint, and their cardinalities add.

We prove the other two estimates next. The first counts exterior edges that survive host assignment. The second counts internal bases that may be left without a host after that choice. The order matters: second-phase lists exclude the links actually used by the first phase.

First phase: color selection and incompatibility losses. Write $H = G - R$, $w = \omega(H)$ and $c = \max\{p, d_{\max}(H) + 1\}$, where $d_{\max}(H)$ is the maximum degree. Vizing supplies a proper coloring with at most c colors. Among colorings on this palette, choose one minimizing the sum of squared class sizes. If two classes differed by at least two edges, their union would contain an alternating path with one more edge of the larger color; switching colors along it would decrease the sum of squares. Thus class sizes differ by at most one, and each class has at most $t = \lceil m/c \rceil$ edges.

Retain the p largest classes. If they contain m_0 edges in total, comparing their average size with the average over all classes gives

$$m_0 \geq \frac{p}{c}\, m. \tag{5.5}$$

Take a perfect elimination ordering ending in R and orient every exterior edge toward its later endpoint. If $u \to v$, the later neighbors of u form a clique, so every neighbor of u in R is also a neighbor of v. Consequently, the invalid hosts for uv are exactly the vertices of R nonadjacent to u. If their number is a_u, vertex u contributes at most $(w-1)a_u$ invalid incidences: it has at most $w-1$ later exterior neighbors. Summing and using $\sum_{u \notin R} a_u = A$ gives at most $(w-1)A$ invalid incidences.

Enumerate the retained classes and the root cyclically. Across the p shifts, each class receives each host once. The average number of discarded edges is therefore at most $(w-1)A/p$. Some shift retains a number f of edges satisfying

$$f \geq m_0 - \frac{w-1}{p} A \geq \frac{p}{c}\, m - \frac{w-1}{p} A. \tag{5.6}$$

Each retained edge produces a triangle. Within a class the bases form a matching, and distinct classes have distinct hosts; hence the triangles are edge-disjoint. The palette and width bounds supplied by the regularized root are

$$40c \leq 73p, \qquad 40(w-1) \leq 3p. \tag{5.7}$$

Substitution into (5.6) gives $f \geq 40m/73 - 3A/40$. Multiplication by 2920 gives exactly $1600m \leq 2920f + 219A$, the first estimate in Table 2.

Second phase: one or two candidates per factor. Factor K_p into matchings and, if needed, add an empty class to work with p factors. Since $q \geq p$, we can give each factor one candidate. Let $s = \max\{2p - q, 0\}$. Choose s factors with a single candidate and give two to each remaining

factor. The $2p - s$ available positions are assigned injectively to exterior vertices: no candidate belongs to two factors.

For $x \in R$, let a_x count missing links and u_x count links already used by the first phase. Define $d_x = a_x + u_x$. We have

$$\sum_x a_x = A, \quad \sum_x u_x = 2f, \quad a_x \le D, \quad u_x \le 2t, \quad S := \sum_x d_x = A + 2f, \tag{5.8}$$

where $D = \max_x a_x$. The last pointwise bound follows because the matching assigned to x has at most t bases. For the base $e = xy$, let b_e count invalid candidates: one of the links xz, yz is missing or has already been used. The union of these prohibitions gives $b_e \le d_x + d_y$.

The three moment estimates follow by counting over unordered pairs in the root:

$$\begin{aligned} \sum_e b_e &\le (p-1)S, \\ \sum_e b_e^2 &\le (p-2)\sum_x d_x^2 + S^2, \\ \sum_x d_x^2 &\le (D+4t)A + 4tf. \end{aligned} \tag{5.9}$$

The first counts each d_x in the $p-1$ bases containing x. For the second, expand $\sum_{x<y}(d_x+d_y)^2$: squares occur $p-1$ times, and cross products sum to $S^2 - \sum_x d_x^2$. For the third, use term by term

$$(a_x + u_x)^2 \le Da_x + 4ta_x + 2tu_x,$$

and apply the sums in (5.8). Thus every term in (5.9) comes from actual missing or occupied incidences.

Now average over the uniform choice of the s single-candidate factors and over candidate injections. A base with one candidate fails with probability b_e/q; with two distinct candidates it fails with probability $b_e(b_e-1)/(q(q-1))$. Its total failure probability is therefore

$$\frac{s}{p}\frac{b_e}{q} + \left(1 - \frac{s}{p}\right)\frac{b_e(b_e-1)}{q(q-1)}. \tag{5.10}$$

Sum and use $(p-1)/p \le 1$, $b_e(b_e-1) \le b_e^2$ and (5.9). Some assignment has a number ℓ of failed bases no greater than the average, so

$$\ell \le \frac{s}{q}(A+2f) + \frac{(p-2)((D+4t)A+4tf) + (A+2f)^2}{q(q-1)}. \tag{5.11}$$

For each base that does not fail, choose one valid candidate. Bases in the same factor are disjoint; bases in different factors use distinct candidates. Moreover, validity excludes every link already used. This simultaneously verifies compatibility within the second phase and with the first-phase packing.

It remains to evaluate (5.11) using the regularization bounds. Let $a = |P|$ denote the size of the reference clique before (5.2). The near certificate gives

$$\begin{gathered} a \ge 1024, \quad \frac{q}{a} \ge \frac{87947}{44352}, \quad \frac{p}{a} \le \frac{101}{100}, \quad \frac{D}{a} \le \frac{1}{3}, \\ \frac{t}{a} \le \frac{1}{256}, \quad \frac{s}{a} \le \frac{1}{48}, \quad \frac{A+2f}{a^2} \le \frac{11}{2000}. \end{gathered} \tag{5.12}$$

These bounds belong to the certificate of the regularized root and its first phase. For example, that certificate gives $p \ge 99a/100$, $m < a^2/400$ and $2000(A+2m) \le 11a^2$. Since $c \ge p$, each

class size satisfies $t \le m/p + 1 < a/396 + 1 \le a/256$, where the last comparison uses $a \ge 1024$. Also $f \le m$, so the mass bound on $A + 2f$ follows from that on $A + 2m$. Thus the quantities in (5.12) belong to the chosen construction; they are not parameters adjusted after counting failures. The bound on s requires the fine estimates in (5.3), not merely the rounded endpoints for p/a and q/a. If $q_P = n - a$, then $p = a - |X| + |Y|$, $q = q_P + |X| - |Y|$, and

$$2p - q = 2a - q_P - 3|X| + 3|Y| \le a/64 + 3a/57344 < a/48. \tag{5.12a}$$

Taking the positive part proves $s/a \le 1/48$. The denominator, in turn, retains the term $q - 1$:

$$\frac{q(q-1)}{a^2} \ge \left(\frac{87947}{44352}\right)^2 - \frac{87947}{44352 \cdot 1024} \ge \frac{393}{100}. \tag{5.13}$$

The middle expression is approximately 3.93008, whereas the bound used is 3.93. The difference is small but positive; we keep the exact rational comparison, without rounding it upward or changing R_0 in the subsequent accounting. To linearize the numerator, use $(A + 2f)^2 \le (11/2000)a^2(A + 2f)$ and $p - 2 \le p$. Writing $Q_0 = 87947/44352$ and $R_0 = 393/100$, the coefficients of A and f are bounded, respectively, by

$$\begin{aligned} \frac{1}{48Q_0} + \frac{(101/100)(1/3 + 1/64) + 11/2000}{R_0} &\le \frac{11}{100}, \\ \frac{1}{24Q_0} + \frac{(101/100)/64 + 11/1000}{R_0} &\le \frac{29}{1000}. \end{aligned} \tag{5.14}$$

These are direct rational comparisons. We obtain $\ell \le 11A/100 + 29f/1000$. Since $A, f \ge 0$, this implies $\ell \le 7A/40 + f/25$, which, multiplied by 200, is the second estimate in Table 2. Thus both contributions, from missing links and from occupied links, have been used on a single physical assignment.

Lemma 5.3

Retaining the stronger estimate in (5.14) for the same partition Q gives

$$|Q| \le B_n(p) - \frac{117}{1825}m - \frac{12687}{20000}A \le B_n(p) - \frac{m}{16} - \frac{A}{2}. \tag{5.15a}$$

In particular, the three accounts in Table 2 also imply the earlier form

$$|Q| \le B_n(p) - \frac{19}{365}m - \frac{253}{500}A \le B_n(p) - \frac{m}{20} - \frac{A}{2}. \tag{5.15}$$

Proof. The second-phase account gives $\ell \le 7A/40 + f/25$. Substitution into the piece identity gives

$$|Q| \le B_n(p) + m - \frac{13}{20}A - \frac{48}{25}f.$$

The first phase gives $f \ge 40m/73 - 3A/40$. Its coefficient in the preceding inequality is negative; this is why the lower bound is used. The resulting coefficients are

$$1 - \frac{48}{25}\frac{40}{73} = -\frac{19}{365}, \qquad -\frac{13}{20} + \frac{48}{25}\frac{3}{40} = -\frac{253}{500}.$$

Since $m, A \ge 0$, $19/365 \ge 1/20$ and $253/500 \ge 1/2$, the second inequality follows.

For (5.15a), return to the estimate before that relaxation, $\ell \le 11A/100 + 29f/1000$, without altering the physical assignment. The piece identity gives

$$|Q| \le B_n(p) + m - \frac{39}{50}A - \frac{971}{500}f.$$

Using $f \geq 40m/73 - 3A/40$ again changes the coefficients to $-117/1825$ and $-12687/20000$. Finally, $117/1825 > 1/16$ and $12687/20000 > 1/2$. This reserve requires neither a new root nor a new packing.

On the other hand,

$$B_n(p) = \frac{(2n+1)^2}{24} - \frac{(6p-2n-1)^2}{24}. \tag{5.16}$$

The integer form of the same identity is (6.3): $6B_n(p) + (n-3p)(n-3p+1) = n(n+1)$. Since the product of consecutive integers is nonnegative, $6B_n(p) \leq n(n+1)$. Now integrality of $B_n(p)$ gives $B_n(p) \leq \lfloor n(n+1)/6 \rfloor = M(n)$. The floor has not been identified with the rational function: the envelope in (5.16) exceeds $n(n+1)/6$ by $1/24$, and that term must not be dropped.

5.4. The witness retained by the proof

The near conclusion supplies, together, the comparator of (4.7)–(4.8), the regularized root, a packing in G and the accounts in Table 2. By (5.15)–(5.16), its completion has cost at most $M(n)$.

In particular, we do not construct a cheap partition in S_C and then repair all its differences from G. Edits serve to locate and calibrate the root. The partition is realized directly in G, and losses are paid by its own families of edges and triangles.

Corollary 5.4 (quantitative geometry of the critical branch). For every sufficiently large chordal graph G, either $F_4(G) < n^2/6 - \eta_0 n^2$, or there are a clique R of the original graph and a comparator core C such that

$$|3|R| - n| \leq \frac{n}{50}, \qquad |3|C| - n| \leq \frac{n}{10^4}, \qquad d_E(G, S_C) \leq \varepsilon_0 n^2, \tag{5.17}$$

and, in the second branch,

$$\left| e(G) - \frac{5n^2}{18} \right| \leq \frac{n^2}{10^4}. \tag{5.18}$$

For (5.17), it suffices to take $n \geq 4 \cdot 10^{12}$ in addition to the threshold of the dichotomy. The actual clique follows from the three regularization ratios used in Theorem 5.0. The bound on C follows from (4.8) and the calibration. If $k = |C| = n/3 + t$, then $|t| \leq n/30000$ and

$$e(S_C) = \binom{k}{2} + k(n-k) = \frac{5n^2}{18} + \frac{2nt}{3} - \frac{t^2}{2} - \frac{k}{2}.$$

The last bound follows by comparing this count with $e(G)$ using (4.7). The constant $5/9$ describes the *asymptotic* density relative to a complete graph; it is not an exact density identity at finite order. The corollary controls sizes, not uniqueness of R or C as vertex sets.

6. Assembly and extremal value

Proof of Theorem B: upper bound

Let N_{far} be the threshold in Corollary 3.5 with $\eta = \eta_0$, and take $N \geq \max\{N_{\mathrm{far}}, 4 \cdot 10^{12}, 6\}$. For a chordal graph G of order $n \geq N$, construct a certified mixed optimum. If $F_4(G) < n^2/6 - \eta_0 n^2$, Corollary 3.5 supplies the partition. Otherwise, Sections 4 and 5 produce the near witness and a completion meeting the target. The two cases exhaust the values of $F_4(G)$, and both outputs are partitions of the same original graph.

The optimum is chosen within the argument. Nor is a proximity parameter left to be chosen separately for each graph: η_0, ε_0 and all thresholds are fixed in advance.

6.1. Unrestricted optimality for complete-split graphs

Let $S = K_k \vee I_h$, with $2 \le k \le h$. Its edges consist of $\binom{k}{2}$ internal edges and kh links. The weight argument of Paper III [4, Corollary 10.2a] assigns weight -1 to the former and $+1$ to the latter.

Every clique contains at most one host. If it contains one host and $s \ge 1$ core vertices, its weight is

$$s - \binom{s}{2} \le 1.$$

If it lies entirely in the core, its weight is negative. Thus every partition $\mathcal{Q}$, without any restriction on clique order, satisfies

$$|\mathcal{Q}| \ge kh - \binom{k}{2}. \tag{6.1}$$

To attain the bound, factor the edges of K_k into $k-1$ perfect matchings if k is even, and into k matchings if k is odd. In the latter case, factor K_{k+1} and remove the added vertex. Since $h \ge k$, each class receives a distinct host. This gives one triangle per core edge, with no shared links. Completing the remaining links by individual edges gives the following total piece count:

$$\binom{k}{2} + kh - 2\binom{k}{2} = kh - \binom{k}{2}.$$

Together with (6.1), this equality proves

$$\mathrm{cp}(S) = c_3(S) = c_4(S) = kh - \binom{k}{2}. \tag{6.2}$$

6.2. The eventual maximum

An identity explains the choice of the extremal core. For $f(n,k) = k(n-k) - \binom{k}{2}$, with subtraction interpreted in the integers,

$$6f(n,k) + (n-3k)(n-3k+1) = n(n+1). \tag{6.3}$$

The product of consecutive integers is nonnegative. Hence $f(n,k) \le M(n)$. To attain the floor, this product must equal the remainder of $n(n+1)$ modulo six, which is zero or two. Thus $n-3k$ can only belong to $\{-2,-1,0,1\}$. Imposing its congruence modulo three gives exactly the sizes

$$\begin{cases} k = r, & n = 3r, \\ k = r \text{ or } r+1, & n = 3r+1, \\ k = r+1, & n = 3r+2. \end{cases} \tag{6.4}$$

Within the complete-split family with $2 \le k \le n-k$, the optimal size is unique except in the second row, where there are two consecutive sizes. This classifies the optimal cores within that family; it does not yet classify all extremal chordal graphs.

For $n \ge 6$, the choice $k = \lceil n/3 \rceil$ always belongs to (6.4) and satisfies $2 \le k \le h = n-k$. The graph $K_k \vee I_h$ is chordal: first eliminate the hosts, whose later neighbors lie in the complete core, and then the core. This is a perfect elimination ordering. By (6.2), this chordal graph attains $M(n)$. Together with the upper bound, this yields the maximum in Theorem B.

6.3. All orders and optimality of the linear term

Proof of Theorem A. Fix a global threshold N from Theorem B and put $b = N^2$. If $n \geq N$, there is already a partition with at most $M(n)$ pieces. If $n < N$, cover each edge separately. The piece count is

$$e(G) \leq \binom{n}{2} \leq N^2 \leq M(n) + b.$$

This also covers $n = 0$. Finally, $M(n) \leq n^2/6 + n/6$; for $n \geq 1$, the constant $C = b + 1/6$ gives the form $n^2/6 + Cn$, and for $n = 0$ the partition is empty. This proves the answer to the problem with no exceptions in the order.

The witness $b = N^2$ depends on the **global** threshold, not only on the near threshold $4 \cdot 10^{12}$. It can now be chosen explicitly. Define the tower of twos by $T(0) = 1$ and $T(j+1) = 2^{T(j)}$, and set

$$h = 4 \cdot 8^5 \cdot 2208^5 \cdot 4500^{105} \cdot (2 \cdot 10^{16})^{105}.$$

The calculation in Appendix C.3 allows the choices

$$N = T(h+7), \qquad b = N^2 \leq T(h+8).$$

Indeed, the far branch with $\eta_0 = 10^{-16}$ requires at most $T(h+7)$, and this number also exceeds the near threshold. For $n < N$, the partition into individual edges has at most $n^2 \leq N^2$ pieces. Finally, $N \geq 4$ allows the integer inequality $N^2 \leq 2^N = T(h+8)$.

This bound is explicit but not practical, and no optimality is claimed. It does not prove $b = 0$, nor replace the tower by a polynomial threshold. The evaluation includes the marked selector's constants: no unspecified rounding threshold remains inside the expression. The identity with the continuous term is still $M(n) = n^2/6 + n/6 - \vartheta_n$, with $0 \leq \vartheta_n < 1$.

Optimality of the quadratic coefficient. The $1/6$ in $n^2/6$ cannot be lowered, even when cliques of arbitrary order are allowed. Indeed, suppose a bound $\mathrm{cp}(G) \leq cn^2 + Cn$ holds for every sufficiently large chordal graph. Apply it to the critical complete-split witness of §6.2. By (6.2) and the definition of $M(n)$,

$$\frac{n^2+n}{6} - 1 \leq M(n) \leq cn^2 + Cn. \tag{6.5a}$$

If $c < 1/6$, the difference $(1/6 - c)n^2$ exceeds $|C|n + 1$ for large n, contradicting (6.5a). The mathematical statement for real coefficients follows by choosing rationals between c and $1/6$, and above C. This differs from optimality of the *linear* coefficient below; neither claim settles whether $b = 0$.

With the quadratic term fixed at $n^2/6$, the coefficient $1/6$ of the linear term is also optimal. Given $c < 1/6$ and a constant B_0, for every sufficiently large order the complete-split witness satisfies

$$\mathrm{cp}(G) = M(n) \geq \frac{n^2}{6} + \frac{n}{6} - 1 > \frac{n^2}{6} + cn + B_0. \tag{6.5}$$

The last inequality holds once $(1/6 - c)n > B_0 + 1$. The passage from rational to real parameters also follows by choosing a rational between c and $1/6$ and another larger than B_0. This distinguishes optimality of the asymptotic term from the additive constant needed for small orders.

The order restriction also clarifies the scope of Theorem A: dropping the restriction on piece order immediately gives the $\mathrm{cp}(G) \leq M(n) + b$ form of the original problem. The reverse implication is not valid for arbitrary bounds: for $n \geq 2$, $\mathrm{cp}(K_n) = 1$, whereas a piece of order at most four covers at most six edges, so $c_4(K_n) \geq \binom{n}{2}/6$. This comparison of parameters does not establish a logical separation between the two universal statements with the particular target $M(n)$.

6.4. The excess of a complete-split partition

Retain the regime $2 \le k \le h$ of Section 6.1. Its weight argument can be read piece by piece. For a clique C of $K_k \vee I_h$, define

$$d(C) = 1 + e_{\mathrm{int}}(C) - e_{\mathrm{cross}}(C),$$

where internal core edges and core–host edges are counted separately. Then every partition $\mathcal{Q}$, without restriction on order, satisfies

$$|\mathcal{Q}| - \left(kh - \binom{k}{2}\right) = \sum_{C \in \mathcal{Q}} d(C). \tag{6.6}$$

Proof. Internal edges sum to $\binom{k}{2}$ over the pieces, and cross edges sum to kh, since coverage is exact. Summing the definition gives (6.6). If C has one host and $s \ge 1$ core vertices,

$$d(C) = 1 + \binom{s}{2} - s = \frac{(s-1)(s-2)}{2} \ge 0;$$

if it has no host, $d(C) = 1 + \binom{|C|}{2} > 0$. No piece consists solely of hosts: they form an independent set, and pieces have at least two vertices. The preceding cases are therefore exhaustive. The defect is zero exactly for core–host edges and triangles with two core vertices and one host. An optimal partition uses only these pieces. More generally, a partition with excess at most t has at most t positive-defect pieces, since each contributes at least one unit to (6.6). Uniqueness of the optimal partition is not asserted: there may be different assignments of the same piece types.

6.5. Integral stability and classification of extremizers

The upper-bound argument retains the stronger near account (5.15a), although Theorem B needs only (5.15). A graph whose optimal partition count is close to $M(n)$ cannot lie in the far branch, where rounding produces a partition well below that target. The retained account therefore converts a small integral deficit into a bound on the graph's exterior edges and missing links. Once the root has been chosen, a separate piecewise identity controls every partition close to the same target; the root is not selected afresh for each partition.

Theorem 6.1 (integral stability). There exist $\gamma > 0$ and a threshold N_{est} such that, if $n \ge N_{\mathrm{est}}$, G is chordal, $0 \le \delta \le \gamma n^2$, and every partition of order at most four has at least $M(n) - \delta$ pieces, then there is a clique R of G for which

$$\big(M(n) - B_n(|R|)\big) + \frac{m}{16} + \frac{A}{2} \le \delta, \qquad d_E(G, S_R) = m + A \le 16\delta. \tag{6.7}$$

Here $m = e(G - R)$ and A is the number of missing links between R and $V(G) \setminus R$.

We explain the proof. Take $\gamma = \eta_0/4$ and apply the dichotomy. In the far branch, RC01 retains at least $\eta_0 n^2/2$ of margin and produces a partition with

$$|Q| < \frac{n^2}{6} - \frac{\eta_0}{2} n^2 \le M(n) - \frac{\eta_0}{2} n^2.$$

This contradicts $|Q| \ge M(n) - \delta$, because $\delta \le \eta_0 n^2/4$. Thus only the near branch can occur. There, the stronger account (5.15a), retained for the same physical witness, gives

$$M(n) - \delta \le |Q| \le B_n(|R|) - \frac{m}{16} - \frac{A}{2},$$

which yields the first inequality in (6.7). Since $B_n(|R|) \le M(n)$ and $A \ge 0$,

$$m + A \le 16\left(\frac{m}{16} + \frac{A}{2}\right) \le 16\delta.$$

Finally, the fact that R is a clique identifies exactly the edits needed to transform G into S_R: delete the m exterior edges and add the A missing links. This proves the equality in (6.7).

The same root controls not only the graph but also its partitions close to the maximum. For any partition Q, even when cliques of order greater than four are allowed, put $a_K = |K \cap R|$ and $b_K = |K \setminus R|$ for each piece K. Define its defect relative to R by

$$d_R(K) = 1 + \binom{a_K}{2} + 3\binom{b_K}{2} - a_K b_K = \frac{(a_K - b_K)(a_K - b_K - 1)}{2} + (b_K - 1)^2. \tag{6.7a}$$

The first term on the right is nonnegative for every integer $a_K - b_K$. Thus $d_R(K) = 0$ exactly for root–exterior edges and triangles with two root vertices and one exterior vertex.

Corollary 6.1a (stability of partitions). Under the hypotheses of Theorem 6.1, the root R can be chosen so that every unrestricted clique partition Q with $|Q| \le M(n) + \tau$, $\tau \ge 0$, has at most $\tau + 48\delta$ noncanonical pieces. The total number of edges in those pieces is at most $10\tau + 480\delta$. Both conclusions use the same root; there is no restriction on piece order.

Proof. Exact edge coverage counts the edges internal to R, exterior to R, and crossing the cut as $\binom{|R|}{2}$, m, and $|R|(n - |R|) - A$, respectively. Summing (6.7a) over the pieces yields

$$\sum_{K \in Q} d_R(K) = |Q| - B_n(|R|) + A + 3m. \tag{6.7b}$$

Each noncanonical piece contributes at least one. If $r = M(n) - B_n(|R|)$, the assumption on Q bounds the total defect by $\tau + r + A + 3m$. The reserve in (6.7) implies $r + A + 3m \le 48\delta$. For the edge count, the local numerical bound $\binom{a+b}{2} \le 10 d_R(K)$ holds whenever $d_R(K) > 0$, where $a = |K \cap R|$ and $b = |K \setminus R|$. The zero-defect pieces are exactly the canonical ones. Summing this inequality over the noncanonical pieces bounds their edges by ten times the same total defect, giving $10\tau + 480\delta$. The root comes from the witness in Theorem 6.1 and does not depend on Q.

Corollary 6.2 (eventual classification). Extremal chordal graphs of sufficiently large order are exactly the complete-split graphs with the core sizes in (6.4).

Proof. Setting $\delta = 0$ in (6.7) forces $m = A = 0$, so $G = S_R$. The arithmetic identity

$$6B_n(k) + (n - 3k)(n - 3k + 1) = n(n + 1) \tag{6.8}$$

then classifies the sizes attaining $M(n)$. Consequently, for sufficiently large n,

$$\mathrm{cp}(G) = M(n) \iff c_4(G) = M(n) \iff G \text{ is complete-split with an optimal core,} \tag{6.9}$$

where the core size is r if $n = 3r$, r or $r+1$ if $n = 3r+1$, and $r+1$ if $n = 3r+2$. The equivalence uses the unrestricted lower bound in Section 6.1; it therefore classifies extremizers for both cp and c_4.

6.6. Fixed-defect stability on one root

Theorem C supplies a sharp value. To retain the geometry behind that value, partition the vertex set as $V = C \sqcup D \sqcup H$, where C is a clique of G, $|D| = s$, and $2 \le |C| \le |H|$. The associated **defect template** is

$$T(C, D, H) = (K_C \sqcup I_D) \vee I_H.$$

Only the clique condition on C is imposed in the original graph: D and H need not be independent there. Set $k = |C| + s$. The template has baseline

$$B_{s,n}(k) = k(n-k) - \binom{k-s}{2}. \tag{6.10}$$

A canonical piece is an edge between $C \cup D$ and H, or a triangle with two vertices in C and one in H. Write $N_{\mathrm{bad}}(Q)$ for the number of other pieces and $E_{\mathrm{bad}}(Q)$ for the sum of their edge counts. The latter counts distinct edges, because Q is a partition.

Theorem C' (fixed-defect same-root stability). For each fixed integer $s \geq 0$, there are explicit $\gamma_s > 0$ and N_s^{stab} with the following property. Put

$$\epsilon_s = \frac{1}{10^{41}(s+1)^8}, \qquad A_s = \max\{40000(s+1)^3,\, 2/\epsilon_s\}. \tag{6.11}$$

If $n \geq N_s^{\mathrm{stab}}$, $\mathrm{rsd}(G) \leq s$, and

$$c_4(G) \geq Q_s(n) - \delta, \qquad 0 \leq \delta \leq \gamma_s n^2,$$

then there is a single root (C, D, H) as above such that

$$\begin{aligned} d_E(G, T(C,D,H)) &\leq A_s\delta, \\ 0 \leq Q_s(n) - B_{s,n}(k) &\leq (1 + 4A_s)\delta. \end{aligned} \tag{6.12}$$

For this same root, every unrestricted clique partition Q with $|Q| \leq Q_s(n) + \tau$, $\tau \geq 0$, satisfies

$$\begin{aligned} N_{\mathrm{bad}}(Q) &\leq \tau + (1 + 7A_s)\delta, \\ E_{\mathrm{bad}}(Q) &\leq 10\tau + 10(1 + 7A_s)\delta. \end{aligned} \tag{6.13}$$

Both δ and τ may be real. The root is chosen before Q and τ. Figure 3 summarizes the construction and its two conclusions.

The maximum in (6.11) equals $2 \cdot 10^{41}(s+1)^8$: its second term dominates the terminal constant $40000(s+1)^3$ for every $s \geq 0$. The coefficients are not asserted optimal. At $s = 0$, the specialized chordal argument remains preferable: it gives 16, 48, and 480, rather than the large constants in (6.11).

Proof. We first explain the retained estimate at the minimum-degree terminal, then the passage to an arbitrary graph in the class. In the normalized presentation of Appendix E, let a count missing S–H links, $m = e(G[H])$, $t_W = |W|$, and

$$E = e(G[S]) - \binom{|S| - s}{2} \geq 0.$$

The constructor retains a nonnegative residual credit R_*. Extraction of an actual clique $C \subseteq S$ gives $|S| = |C| + s + r$, with $r \geq 0$. The retained accounts are

$$\begin{gathered} a + \frac{m}{2000(s+1)^2} + R_* \leq \delta, \qquad nt_W \leq 800(s+1)^2 R_*, \\ E \leq R_* + nt_W, \qquad rn \leq 16sE, \\ d_E(G, T(C,D,H)) \leq a + m + E + 2rn + 2nt_W. \end{gathered} \tag{6.14}$$

Here D, H in the last line are the final root classes, not necessarily the intermediate host class. Appendix E.4 derives all five accounts: reducing the first-phase palette retains a positive multiple of m, the strict exceptional-vertex budget pays nt_W, and an actual clique is extracted

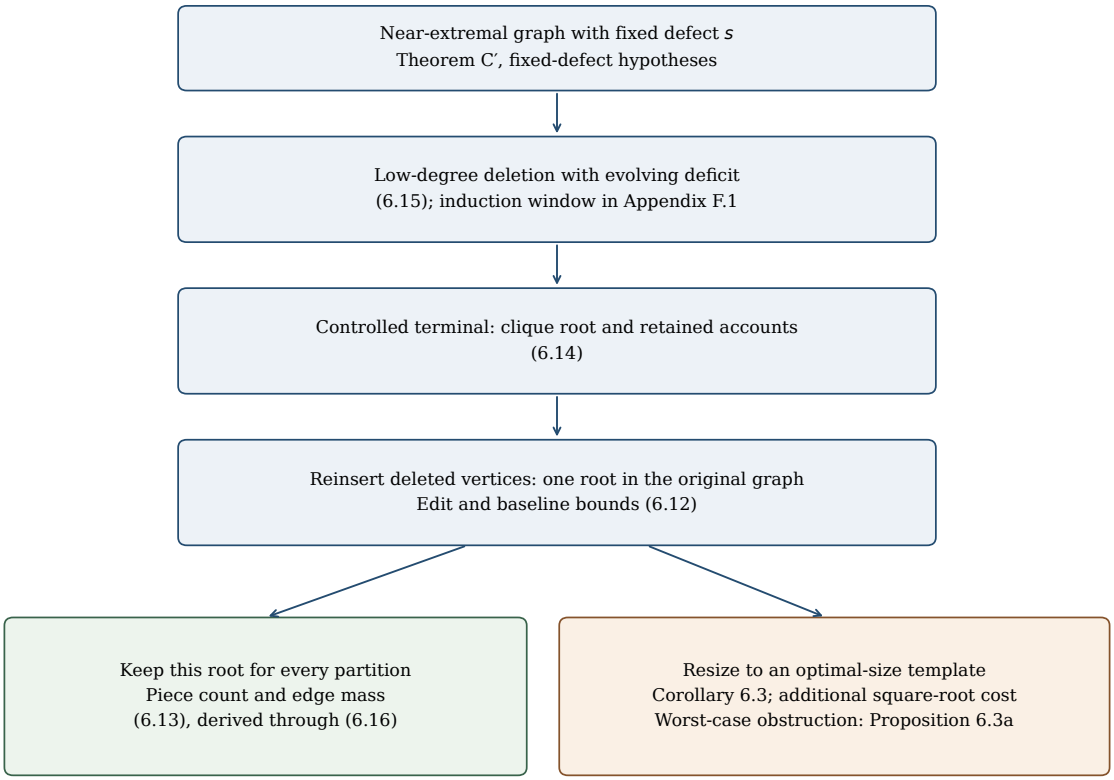


Figure 3: Theorem C': low-degree deletion reaches the controlled terminal, and reinsertion transports its root back to the original graph. The retained accounts then control every partition on that same root. Resizing to an optimal template is a separate step (Corollary 6.3), with the square-root cost illustrated by Proposition 6.3a. The diagram is schematic; Appendix F.1 supplies the induction window.

from S before the final template is formed. In particular, the two charges rn have different origins: reducing the clique size in the interior count, and moving the remaining vertices to the host class.

Substituting the middle bounds into the last gives

$$d_E(G,T) \le a + m + \big(1 + 32s + 800(3+32s)(s+1)^2\big)R_*.$$

The coefficient of R_* is at most $38000(s+1)^3$. Since $a + m/[2000(s+1)^2] + R_* \le \delta$, it follows that $d_E(G,T) \le 40000(s+1)^3\delta$. This explains the terminal constant in (6.11), rather than introducing it as an unexplained loss.

Low-degree deletion extends the conclusion to all graphs of rooted defect at most s. At current order t, delete a vertex of degree less than $(1/3 - \epsilon_s)t$, and put

$$u = Q_s(t) - Q_s(t-1), \qquad \delta' = \delta + \deg(v) - u. \tag{6.15}$$

Every order-four partition of the child has cost at least $Q_s(t-1) - \delta'$: otherwise restoring the incident edges contradicts the hypothesis on the parent. Theorem C for the child implies $\delta' \ge 0$. Also $u \ge (t-1)/3$, so

$$\delta' \le \delta - \epsilon_s t + \frac{1}{3}.$$

Reinserting the vertex into the host class of the child's template changes at most $t-1$ edges. The coefficient $A_s \ge 2/\epsilon_s$ pays this cost. The induction uses the window $\delta \le \gamma_0 t(t - N_0)$, with $\gamma_0 \le \epsilon_s/4$, rather than repeatedly assuming a quadratic window of unchanged radius. Above $2N_0$, the latter follows with $\gamma_s = \gamma_0/2$. The exact parameter choices are in Appendix F. This proves the first line of (6.12).

For the same root, a partition of the template can be repaired to a partition of G with order at most four and cost at most $B_{s,n}(k) + 4d_E(G,T)$. Comparing with the lower bound $Q_s(n) - \delta$ proves the upper baseline deficit in (6.12); the lower one is the integer comparator inequality.

Finally, piece accounting relative to $C \cup D$ and H gives

$$\begin{aligned} N_{\mathrm{bad}}(Q) &\le |Q| - B_{s,n}(k) + 3d_E(G,T), \\ E_{\mathrm{bad}}(Q) &\le 10\big(|Q| - B_{s,n}(k) + 3d_E(G,T)\big). \end{aligned} \tag{6.16}$$

The numerical defect is the one in (6.7a). A zero-defect triangle that is not canonical must use an extra edge within $C \cup D$, outside C; these triangles are charged to distinct such edges. Positive-defect pieces satisfy $\binom{|K|}{2} \le 10d(K)$. Summation gives (6.16), and substitution of (6.12) proves (6.13). For real deficits, integer partition costs permit replacement of δ by $\lfloor\delta\rfloor$ in the rational proof; monotonicity then restores the displayed real bounds.

Corollary 6.3 (optimal-size comparison and equality). Under the hypotheses of Theorem C' there is an optimal-size extremal template T_* with

$$d_E(G,T_*) \le A_s\delta + n\sqrt{(1+4A_s)\delta}. \tag{6.17}$$

At $\delta = 0$, for sufficiently large n,

$$\mathrm{cp}(G) = Q_s(n) \iff c_4(G) = Q_s(n) \iff G \cong (K_{k-s} \sqcup I_s) \vee I_{n-k}, \tag{6.18}$$

where k is an integer nearest to $(2(n+s)+1)/6$.

Proof. Let $\mathcal{K}_{s,n}$ be the set of these nearest integers. The discrete comparator parabola implies

$$\mathrm{dist}(k, \mathcal{K}_{s,n})^2 \le Q_s(n) - B_{s,n}(k). \tag{6.19}$$

Moving d vertices between the core and host classes, while leaving D fixed, changes at most nd edges. Apply (6.19) and (6.12), then the triangle inequality for edit distance, to obtain (6.17). The resized core need not remain a clique of G; the actual-clique assertion belongs to the original root in Theorem C'. The set $\mathcal{K}_{s,n}$, rather than one predetermined maximizer, is essential when $n + s \equiv 1 \pmod 3$.

If $\delta = 0$, (6.12) already gives $G = T(C, D, H)$ and equality of its baseline with $Q_s(n)$. This proves the forward classification for c_4. Since $\mathrm{cp} \le c_4 \le Q_s(n)$, it also covers equality for cp. The unrestricted lower witness in Appendix E gives the converse. Thus the classification recovers the equality statement of [15, Theorem 1.1], with the additional order-four upper bound.

Proposition 6.3a (a square-root obstruction). Fix $s \ge 0$. Let q, d be integers with $2s + 2 \le q$ and $1 \le d \le q/2$, and put $n = 3q - s$. The graph

$$G_{s,q,d} = (K_{q+d-s} \sqcup I_s) \vee I_{2q-s-d}$$

has rooted simplicial defect at most s and satisfies

$$\mathrm{cp}(G_{s,q,d}) = c_4(G_{s,q,d}) = Q_s(n) - \delta_d, \qquad \delta_d = d^2 + \binom{d}{2} = \frac{3d^2 - d}{2}. \tag{6.20}$$

For every optimal-size template T on the same vertex set, with any labelling,

$$d_E(G_{s,q,d}, T) \ge \frac{nd}{4} \ge \frac{n\sqrt{\delta_d}}{8}. \tag{6.21}$$

Proof. The rooted-defect argument in Appendix E.3 does not require an optimal core size: it applies to $G_{s,q,d}$ and gives $\mathrm{rsd}(G_{s,q,d}) \le s$. The clique core has at least two vertices and no more vertices than the host set. The comparator construction therefore gives a partition of order at most four with cost $B_{s,n}(q + d)$. The weight argument of Appendix E.3 applies at this nonoptimal core size as well: every unrestricted partition has at least that many pieces. Subtracting this baseline from $Q_s(n)$ gives (6.20).

Since $n + s = 3q$, the unique optimal combined-core size is q. Every optimal template consequently has the same edge count, independent of its labelling. Counting core edges and cross-links gives

$$2\big(e(G_{s,q,d}) - e(T)\big) = d(4q - 4s - d - 1).$$

In particular,

$$4\big(e(G_{s,q,d}) - e(T)\big) - nd = d(5q - 7s - 2d - 2) \ge 0.$$

For the last inequality, write the second factor as $4(q - 2s - 2) + (q - 2d) + s + 6$. An edge edit changes the total number of edges by at most one, so $d_E(G_{s,q,d}, T) \ge e(G_{s,q,d}) - e(T) \ge nd/4$. Finally, $\delta_d \le 2d^2 \le 4d^2$, proving (6.21). □

Taking $d = 1$ gives exact deficit one and unbounded distance to the entire optimal-size family as q grows. Thus no coefficient depending only on s can replace (6.17) by a bound linear in δ. These examples belong to any fixed positive near-extremality window once their order is sufficiently large. More generally, (6.21) exhibits the square-root scale along the displayed family of deficits. This is a worst-case obstruction, not a pointwise lower bound for every graph or every real deficit, and it does not claim optimal constants. It is consistent with Theorem C': each example is itself a template, but its core is not of optimal size.

6.7. Uniform bounds and sublinear defect

A fixed-s threshold cannot be applied to an arbitrary sequence $s = o(n)$: it may grow much faster than the order. We instead retain a bound uniform in the defect before invoking RC01.

Proposition 6.4 (finite fractional bounds). If $\mathrm{rsd}(G) \le s \le n$, then, at every order including zero,

$$F_4(G) \le M(n-s) + ns. \tag{6.22}$$

On the band $4s \le n$, the stronger bound is

$$F_4(G) \le M(n-s) + ns - \binom{s+1}{2}. \tag{6.23}$$

For every $\varepsilon > 0$, one threshold $N(\varepsilon)$, independent of s, makes the corresponding bound valid for $c_4(G)$ after adding εn^2, whenever $n \ge N(\varepsilon)$ and the respective band condition holds.

The signed-star argument follows [15, Lemma 3.1]; it is cited as a method used here, not as a new localization principle. Our calculation uses $L = 4$, retains the capped exception charge $\sum_{j<q} \min(s, j)$, and bounds the result by the integer comparator rather than rounding down a fractional inequality. Appendix F describes the two star estimates and the small-order check. For the integral conclusion, choose the RC01 threshold from ε first. Completion of the rounded packing costs at most $F_4(G) + \varepsilon n^2$; insert (6.22) or (6.23). The choice of N never depends on the graph or on s.

Theorem 6.5 (sublinear defect and a root for all partitions). Let $n_j \to \infty$, with no monotonicity assumption, and let G_j have order n_j and rooted defect at most $s_j = o(n_j)$. There are partitions Q_j of order at most four such that

$$\max\{0, |Q_j| - n_j^2/6\} = o(n_j^2). \tag{6.24}$$

Suppose in addition that, eventually,

$$c_4(G_j) \ge n_j^2/6 - \delta_j, \qquad \delta_j = o(n_j^2).$$

The deficit sequence may be taken rational. There are cliques R_j of the original graphs and nonnegative errors $e_j = o(n_j^2)$ such that

$$|R_j| = n_j/3 + o(n_j), \qquad d_E(G_j, S_{R_j}) = o(n_j^2), \tag{6.25}$$

and, for every j, every unrestricted clique partition Q and every rational τ with $|Q| \le M(n_j) + \tau$,

$$N_{\mathrm{bad}}(Q) \le \tau + e_j, \qquad E_{\mathrm{bad}}(Q) \le 10\tau + 10e_j. \tag{6.26}$$

Here the canonical pieces are those of Corollary 6.1a relative to R_j. Neither the root nor e_j depends on Q or τ.

Proof. Apply the uniform integral bound of Proposition 6.4 to minimum order-four partitions. For each fixed accuracy, s_j/n_j is eventually small and n_j exceeds its fixed rounding threshold. This proves (6.24); it is an upper-error assertion, not an equality $|Q_j| = n_j^2/6 + o(n_j^2)$ for sparse graphs.

Uniform localization gives, for each accuracy, an actual clique with a small size error and small exterior-and-missing mass. To select one clique independently of the accuracy, minimize over all cliques R of G_j the nonnegative score

$$\mathcal{S}_G(R) = n\big||R| - n/3\big| + e(G - R) + A_R + \big(M(n) - B_n(|R|)\big), \tag{6.27}$$

where A_R counts missing root–exterior links. The family of cliques is finite and nonempty (it includes the empty clique). The localization estimate and the comparator parabola show that the minimum score is $o(n_j^2)$. Each nonnegative summand is bounded by the score, proving (6.25). Put $e_j = 3\mathcal{S}_{G_j}(R_j)$. The identity (6.7b) and the edge-mass inequality used in (6.16) prove (6.26) for all partitions simultaneously. For uniform structural localization, the proof first makes a small chordal edit using the removal development of [22], transfers integral stability, and recovers an actual clique of the original graph. Appendix F.3 gives the losses. This is not an application of Theorem C' with a growing s.

Corollary 6.6 (exact bound, uniform in the defect). Let $T(0) = 1$, $T(j+1) = 2^{T(j)}$, as in §6.3, and define

$$\begin{aligned} c_{\mathrm{poly}} &= 4 \cdot 17664^5 \cdot 9000^{105} + 3006, \\ c_{\mathrm{far}} &= c_{\mathrm{poly}}(128 \cdot 10^{82})^{105}, \\ c_T &= h_{\mathrm{iter}} + c_{\mathrm{far}} + 841780, \qquad P(s) = c_T(s+1)^{1680}, \end{aligned} \tag{6.28}$$

where $h_{\mathrm{iter}} = h$ is the explicit integer defined in §6.3. If $\mathrm{rsd}(G) \le s$ and $n \ge T(P(s))$, then $c_4(G) \le Q_s(n)$. More generally, $n \ge T(P(S))$ suffices simultaneously for every $s \le S$.

For $n \ge T(P(0))$, the finite set below is nonempty, and we may put

$$s_{\max}(n) = \max\{S \in \mathbb{N} : S \le n,\ T(P(S)) \le n\}. \tag{6.29}$$

Then the exact bound holds simultaneously for all $s \le s_{\max}(n)$. The lower restriction on n is essential to this formulation: below it, the defining predicate may have no solution, so returning zero by convention would not imply the bound.

Proof. The threshold comparison gives $F_s \le T(P(s))$ for the explicit threshold in Appendix E.2. Its numerical account combines the uniform far bound (C.4), the chordal-edit threshold and the finite assembly costs. Apply Theorem C at F_s. Both P and T are nondecreasing, so the same threshold for S covers all $s \le S$. Finally, the defining predicate for $s_{\max}$ holds because the set contains zero. □

This is a uniform exact band, not an exact bound for every sublinear defect sequence. Its height is polynomial in s, but the threshold itself is a tower. No assertion of numerical practicality is intended. It bounds the threshold for Theorem C, not automatically the stability threshold N_s^{stab} in (F.1).

7. Formalization and reproducibility

The formal companion [24] records the statement correspondence, dependency scope and reproducibility evidence for the unchanged Lean/Mathlib v4.28.0 source freeze piv-v12-fb459343d234. The recorded checks of the selected declarations use only the standard foundational axioms. The companion distinguishes the main proof library from the separate bounded-clique-gap annex, and formal verification from mathematical and editorial review. The present revision changes exposition only.

8. Comparison of mechanisms and scope

Papers I–III [2–4], in English and Spanish, are collected in the Zenodo v3 deposit dated August 23, 2026, identified by version DOI 10.5281/zenodo.22064657. The dates given in [5] and [15] are September 8 and September 15, 2026, respectively. They are recorded to identify the versions compared; this chronology alone establishes neither mathematical priority nor dependence between proofs.

The general account (1.3a) allows piece families to be compared without identifying the proofs. The following table separates the model used from the conclusion obtained. Here $L \geq 4$ is fixed before the graph, and zero-gain K_2 pieces are added to complete the remaining edges.

Development	Positive-gain pieces	Bounded parameter	Model margin
Preprint [5]	K_3, K_4 in its final construction	c_4	$\Delta_{\{3,4\}}$
Okechukwu [15]	$K_3, \dots, K_L$ in the approximation used for localization	q_L in Lemma 3.4; cp in Theorem 1.1	$\Delta_{\{3,\dots,L\}}$ for the auxiliary model
This work	K_3, K_4	c_4	$\Delta_{\{3,4\}}$

Table 3. Instances of the general budget. q_L is the notation in [15] for the minimum with pieces of order at most L. Margins are compared on the same graph; (1.3b) gives a weak inequality, not a universal strict separation.

All three conclusions can be expressed through (1.3a), each in its own family. Agreement of that accounting identity is not an equivalence of mechanisms, nor does it allow a partition with large pieces to be replaced by one of order at most four. With respect to this restriction, [5] and our conclusion agree; comparison with [15] must keep its larger family visible.

Identity (1.3) allows results to be compared without identifying their proofs. Preprint [5, §§2 and 9] uses fractional-to-integral transfer in the regime with margin. In the critical regime it uses its construction around a root [5, §3], regularization and local stability [5, §§4–5], and first-entry localization [5, §8]. Its eventual bound theorem [5, §9] produces pieces of order at most four. If the K_2 pieces are removed from that partition and we retain as $\mathcal{P}$ the family of K_3, K_4 pieces, Lemma 2.1 gives

$$W^*(G) - g(\mathcal{P}) = \big(e(G) - g(\mathcal{P})\big) - F_4(G) \leq M(n) - F_4(G) = \Delta(G). \tag{8.1}$$

Thus its conclusion satisfies the same budget. This accounting observation is not a formal adapter between its development and our structural witness.

Our proof also separates two cases, but performs the critical step through the descent of Section 4 and the assignment of one or two candidates per factor in Section 5. It retains local stability; it does not claim that every stabilization step can be removed. What it avoids is selecting a first-entry index. Figure 4 locates the differences without portraying the far regime as a stage preceding the critical one.

Component	Preprint [5]	This work
Quadratic margin	Transfer, §§2 and 9	Mixed rounding, §3
Copy path	Adaptation of Paper II, §6	Mixed transport and selection, Lemma 4.1
Localization	First entry, §8	Descent from the terminal graph, §4.2
Near construction	List coloring, §3	Balanced classes and candidates, §5.3
Structural accounting	Local stability, §5	Retained witness and Theorem 6.1
Extremal conclusion	Eventual bound, §9	Theorem B and Corollary 6.2

Table 4. Comparison of steps and source locations. This is not a table of bibliographic priority, nor does it assert that the antecedents of each step belong exclusively to one development.

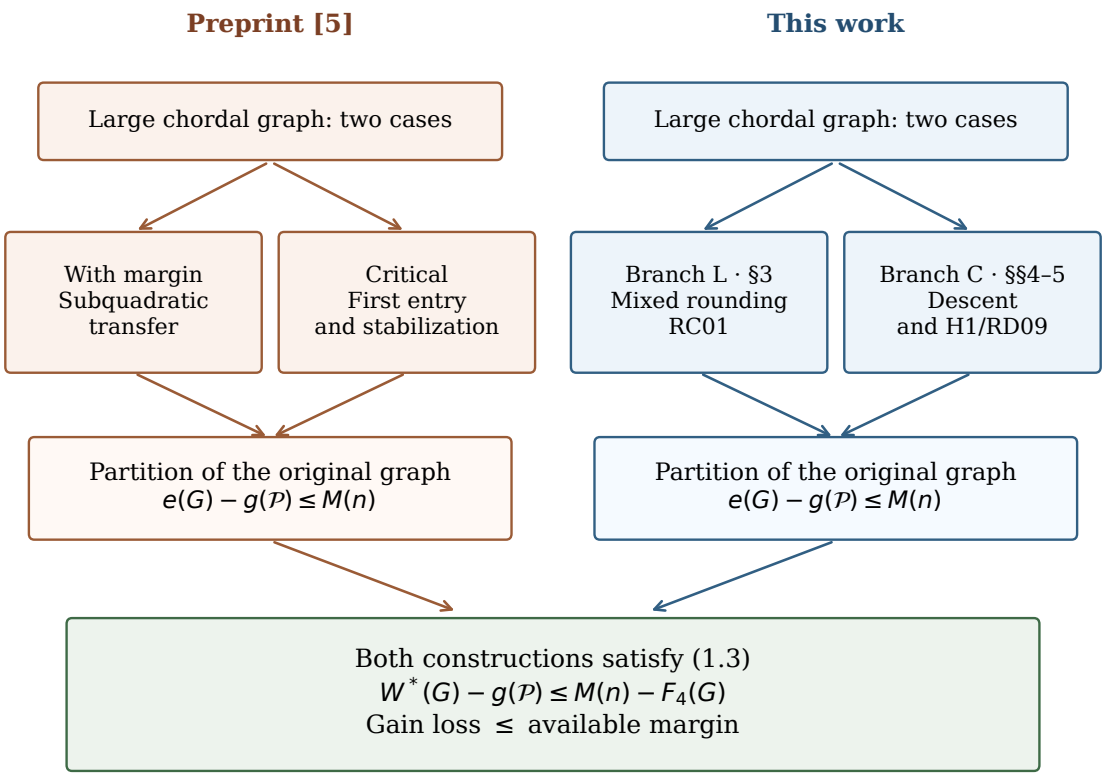


Figure 4: Comparison of mechanisms. Preprint [5] and this work separate two alternative cases and construct a partition satisfying (1.3). Arrows indicate implications toward the conclusion, not passage from one regime to the other. The source locations for the left column are given in Section 8.

The relation to the series is documented in preprint [5] itself. Its introduction cites the fractional bound of Paper II and the split result of Paper III; its Section 6 credits Paper II for the copy scheme, class selection and split terminal, and develops the mixed adaptation. This provenance does not imply that every step of [5] is a formal consequence of the series, just as sharing these antecedents does not identify our two critical mechanisms.

8.1. The bounded simplicial defect antecedent

Okechukwu [15, Theorem 1.1] establishes the fixed-defect eventual maximum, an all-orders additive bound for cp, and the classification of equality graphs. The graph class, $Q_s(n)$, and the attaining family coincide with ours, and we use its rooted-defect definition. Theorems C and C' imply all three clauses: the upper bound holds with pieces of order at most four; the finite remaining range is absorbed into an additive constant; and zero deficit gives the unrestricted equality classification. Neither the extremal formula nor the equality family is presented as new. This comparison does not assert that the method of [15] cannot yield the stronger piece-size bound.

Component	Okechukwu [15]	This work
Localization	Signed dual, Theorem 3.3	Descent by copies, §4
Transfer	Haxell–Rödl, joint templates, Lemma 3.4	Two-quota mixed selection, §3 and Appendix C
Critical construction	List coloring and exceptions, §§4–5	Regularized root and candidates, §5

Table 5. Comparison with the chordal proof in §§3–5; [15] is not identified with [5]. The fixed-defect and sublinear extensions in §§6.6–6.7 also use the localization and removal developments specified in Appendix F. The table is not a claim that all parts of the enlarged paper use descent by copies.

The cutoff in [15] needs care. Theorem 3.3 chooses L as a function of the localization accuracy, and the stability argument used by Proposition 5.4 and Theorem 1.1 applies that cutoff through Lemma 3.4. It does not set $L = 4$. The explicit choice $L = 4$ following the proof of [15, Theorem 1.3] proves its separate approximation (1.5), not the exact bound in Theorem 1.1. Thus our fixed order-four conclusion strengthens the stated exact upper bound; this is not a claim that every possible adaptation of the method in [15] needs larger cliques.

For the auxiliary model in Table 3, [15, Lemma 3.4] gives $q_L(G) \le q_L^*(G) + \zeta n^2$. Its exact fractional partition is equivalent to the gain model of §1.1, so $q_L^* = F_{\{3,\dots,L\}}$. Algebraically, this estimate meets the partition budget if $\zeta n^2 \le M(n) - q_L^*(G)$. This conditional calculation is not a description of the final construction in [15, Theorem 1.1]: that proof uses Lemma 3.4 through structural localization, then a minimum-counterexample argument and the triangle constructions of [15, §§4–5]. The auxiliary cutoff L should not be read as the order of pieces in that final construction.

Theorem C' gives linear edit stability towards a defect template whose core size is not prescribed to be optimal. The optimal-family comparison relevant to [15, Corollary 5.5] is instead (6.17), which retains its square-root term. For fixed s and $\delta = \alpha n^2$, the normalized bound is $A_s\alpha + \sqrt{(1 + 4A_s)\alpha}$, tending to zero with α. This gives an explicit upper bound towards the quantitative question in [15, §7], under the weaker near-extremality hypothesis on c_4: the corresponding lower bound on cp implies it, since cp $\le c_4$. Proposition 6.3a gives a matching square-root scale along an explicit family, for each fixed s. It does not establish optimal constants or optimal dependence on s.

Theorem 6.5 recovers the structural conclusion of [15, Theorem 1.3] and adds simultaneous partition control. Proposition 6.4 together with (6.24) recovers its approximation (1.5) with a conclusion stated for c_4, and retains the finite expression $M(n-s)+ns-\binom{s+1}{2}$ in the band $4s \le n$. The proof of (1.5) in [15] already chooses $L=4$; that choice is not a new ingredient here. Our signed-star argument is explicitly attributed to [15, Lemma 3.1]. The additions here are the retained finite profile and its formal certification. These conclusions do not give linear distance to the optimal-size family, uniform linear stability constants for growing defect, or a replacement for [15, §6].

The term $n/6+1/24$ in the proof of [15, Theorem 3.3] also occurs in our calibration: in both cases $(2n+1)^2/24 = n^2/6+n/6+1/24$. It comes from the continuous quadratic envelope; by itself it is not the floor error in $M(n)$. The integer profile in Appendix D.3 also retains that floor. The optimal cores in [15, Corollary 1.2], described by the integers nearest to $(2n+1)/6$, agree with (6.4), including the tie when $n \equiv 1 \pmod 3$.

The additive form for all orders does not by itself distinguish the results: an eventual bound $c_4 \le M$, such as the one obtained by the construction in [5], implies $c_4 \le M+b$ by absorbing the remaining finite range, exactly as in §6.3.

Quantitative defect control also occurs in [5]: its strict root-regularization lemma bounds the constructed cost by $B_n(p)-m/9-A/2$. We therefore do not present the existence of a linear defect estimate, or the constant 16, as an improvement over that local estimate. The conclusions emphasized here are the stated fixed-defect extension, simultaneous control of unrestricted partitions on one root, and the verified quantitative interfaces.

We also record the project *Clique Partitions of Split Graphs* by Henderson, Koerts, Roberge, Spirkl and Whitman, announced as in preparation [16]. When consulted on 27 September 2026, the page linked no manuscript; accordingly, no results are attributed to it and no overlap is asserted.

Appendix A. Scope of the conclusions

No fractional-to-integral gap $O(n)$ is proved for every chordal graph. Nor is every proof asserted to require copies, regularization or a particular constructor. The eventual bound does not settle whether $b=0$ in Theorem A: that equality would retain, not eliminate, the $n/6$ term contained in $M(n)$.

Appendix B. Complementary construction tools

B.1. Budgeted compatible absorption

The near construction is not the only interface that converts available resources into gain. A perfect matching on an even set U, together with a common host z, absorbs all links zu through triangles: each pair uses two of these links and one base in $G[U]$.

Suppose

$$d_{G[U]}(v) \ge |U|/2 + t \qquad (v \in U), \tag{B.1}$$

with t a nonnegative integer. The infrastructure of Paper III [4] and its matching contribution [9] produce $t+1$ edge-disjoint perfect matchings. Deleting a matching reduces each degree by exactly one, so the degree condition continues to permit the next matching until all these rounds are completed.

For a nonnegative cost $b(u,v)$, averaging over this family selects a matching, represented by an

involution f, with

$$\sum_{u\in U} b(u, f(u)) \le \frac{1}{t+1} \sum_{u\in U} \sum_{v\in U} b(u, v). \tag{B.2}$$

The sum on the left is oriented by vertices. If it is interpreted as a cost per pair, the convention must be adjusted; changing representation does not justify omitting a factor of two.

Now let $\mathcal{P}$ be a prior packing and suppose none of its covered edges has both endpoints in $U \cup \{z\}$. If z is adjacent to all of U, the selected triangles together with $\mathcal{P}$ form a single packing, and

$$g(\mathcal{P} \cup \mathcal{A}) = g(\mathcal{P}) + |U|. \tag{B.3}$$

Indeed, there are $|U|/2$ new triangles, each of gain two. Their bases are disjoint, their links to the host do not repeat, and the free-resource hypothesis guarantees compatibility with $\mathcal{P}$.

Selection and compatibility hold for **the same** certificate. If a budget B_{abs} satisfies

$$\sum_{u\in U} \sum_{v\in U} b(u, v) \le (t+1) B_{\mathrm{abs}},$$

there is a matching whose physical union with $\mathcal{P}$ is a single packing, increases the gain by exactly $|U|$, and has oriented cost at most B_{abs}. First choose the witness in (B.2); then apply compatibility to that witness, since it holds for any matching under the stated free-resource condition. This order avoids combining two different existential choices. The construction requires U to be nonempty and even, as well as the degree condition, adjacency of z and the free resources already specified.

This mechanism does not automatically supply U, z and the free resources in every chordal graph. It provides another physical construction when those data are already available; that interface should not be confused with a second universal proof.

B.2. Balanced reserves

For every finite graph F and every real fraction $0 \le \theta \le 1$, there is $R \subseteq E(F)$ such that

$$|d_R(v) - \theta d_F(v)| \le 2 \qquad (v \in V(F)). \tag{B.4}$$

This is a specialization of Beck–Fiala. To see it, write each edge as its two-element endpoint set and assign it weight θ. The incidence matrix has two ones per column. The Beck–Fiala rounding used in the infrastructure of Paper III selects integral columns and changes each row sum by at most two. The fractional row sums are $\theta d_F(v)$; the integral ones are $d_R(v)$. Passing from edges to their two-element endpoint sets is injective and preserves these degrees.

The result controls the distribution of a reserve, but does not guarantee that a subsequent packing avoids it. It is an auxiliary tool, not another proof of the main theorem.

B.3. Completion with four hosts

Let H be a subgraph of G with maximum degree at most three. Suppose four distinct vertices z_0, z_1, z_2, z_3 are given, none an endpoint of an edge of H, each adjacent in G to all these endpoints. There is a packing of $|E(H)|$ triangles covering all bases of H with gain $2|E(H)|$.

Vizing colors the edges of H with four colors. Each class is a matching; assign to class i the host z_i. Bases are disjoint between classes, hosts are distinct and lie outside the bases, and each host link is used once within its class. The triangles are therefore edge-disjoint and their gain is counted exactly.

The existence of the four hosts is an explicit hypothesis. Adding these triangles to a prior packing also requires their bases and links to be free; the result guarantees neither that freedom nor an automatic application to every chordal residue.

Appendix C. Technical proof of mixed rounding

Lemmas 3.2–3.4 and equations 3.2–3.14 complete the proof of Theorem 3.1. Their numbering is retained to facilitate references from branch L.

Lemma 3.2. Bounded-rank nibble with slack, input from Paper III

For every integer $r \ge 2$ and every $\beta > 0$, there exist $\gamma, C > 0$ with the following property. Let U be a finite set, $\mathcal{H}$ a family of nonempty subsets of U of sizes at most r, and $z_H \ge 0$ weights such that

$$\sum_{H \ni v} z_H \le 1, \qquad \sum_{H \supseteq \{v,w\}} z_H \le \gamma \quad (v \ne w). \tag{3.2}$$

There is a subfamily $\mathcal{M} \subseteq \mathcal{H}$ of pairwise disjoint members with

$$|\mathcal{M}| \ge (1-\beta) \sum_{H \in \mathcal{H}} z_H - \beta |U| - C. \tag{3.3}$$

The constants are fixed before $U, \mathcal{H}, z$. No near-unit load or exceptional set is required. This is the slack form of the nibble from Paper III, not a new result of the present work.

Lemma 3.3. Simultaneous selection of total and marked mass

Let $r \ge 2$, $\beta > 0$, and $\epsilon > 0$. There exist $\gamma, C > 0$ such that, for every r-uniform family $\mathcal{H}$ with nonnegative weights satisfying (3.2), and every marked subfamily $\mathcal{A} \subseteq \mathcal{H}$, there is **a single** matching $\mathcal{M}$ satisfying

$$\begin{aligned} |\mathcal{M}| &\ge (1-\beta) \sum_{H \in \mathcal{H}} z_H - \epsilon |U| - C, \\ |\mathcal{M} \cap \mathcal{A}| &\ge (1-\beta) \sum_{H \in \mathcal{A}} z_H - \epsilon |U| - C. \end{aligned} \tag{3.4}$$

Proof. Set $b = \min\{\beta/2, \epsilon/2\}$ and apply Lemma 3.2 with rank $r+1$ and accuracy b; let γ_0, C_0 be its constants. Write $u = \sum_{H \notin \mathcal{A}} z_H$, $a = \sum_{H \in \mathcal{A}} z_H$, and $k_0 = \lceil 1/\gamma_0 \rceil$. Add two disjoint sets of auxiliary vertices, of sizes

$$p = \max\{\lceil u \rceil, k_0\}, \qquad q = \max\{\lceil a \rceil, k_0\}.$$

Extend each unmarked hyperedge in every possible way by a vertex from the first set, assigning weight z_H/p; marked hyperedges use the second set and weight z_H/q. The original loads are unchanged. The auxiliary loads are u/p and a/q, both at most one. A pair of original vertices retains its codegree; an original–auxiliary pair has codegree at most $1/p$ or $1/q$, and a pair of distinct auxiliary vertices has codegree zero. Lemma 3.2 therefore applies with $\gamma = \gamma_0$.

Projecting the resulting matching onto U cannot identify two of its members: otherwise their nonempty original supports would intersect. Projection thus preserves disjointness and cardinality, and leaves at most p unmarked members. If m is the projected cardinality, then

$$m \ge (1-b)(u+a) - b(|U| + p + q) - C_0, \qquad |\mathcal{M} \cap \mathcal{A}| \ge m - p.$$

Now $p \le u + 1 + k_0$, $q \le a + 1 + k_0$, and summing the loads gives $r(u+a) \le |U|$. For the first quota, the term $b(p+q)$ uses at most $b(u+a) + b(2 + 2k_0)$, and $2b \le \beta$. For the second, after subtracting p, the terms depending on $u + a$ are bounded by $2b(u+a) \le b|U|$. Together with $2b \le \epsilon$, both quotas follow with $C = C_0 + 1 + k_0 + b(2 + 2k_0)$.

Lemma 3.4. Physical mixed selector

For $0 < \beta \le 1$ and $\epsilon > 0$, there exist $\gamma, C, D > 0$ such that the following holds for any graph. Let x be a feasible fractional mixed packing, let $t_3 = \sum_T x_T$ and $t_4 = \sum_Q x_Q$, and suppose that $t_3 \ge C$ and

$$\sum_{K:\ e,f \in E(K)} x_K \le \gamma \qquad (e \ne f). \tag{3.5}$$

Then there is a physical packing $\mathcal{P}$ whose gain satisfies

$$g(\mathcal{P}) \ge (1-\beta)(2t_3 + 5t_4 - 6) - 5\epsilon\binom{n+1}{2} - 5D. \tag{3.6}$$

The quantity $\binom{n+1}{2}$ is the size of the formal universe of unordered pairs, including zero-weight diagonal pairs. This bound may be used without identifying it with $e(G)$.

Proof. We work with resources, namely edges. Disjoint triangle supports are grouped in pairs; a pair T, T' receives weight $x_T x_{T'}/t_3$, summing over representations of the same support. The total mass of pairs is at least $(t_3 - 3)/2$: for each triangle, the mass of triangles meeting it is at most three, by the capacities of its three edges. The load of an edge does not increase. The codegree of the pairs is bounded by the original triangle codegree plus $3/t_3$, according to whether the two resources belong to the same member of the pair or to different members.

Together with the supports of K_4, this gives a 6-uniform family. Mark the latter supports. The two families cannot coincide: two triangles contained in the same K_4 share an edge and cannot form a disjoint pair. If γ_0 is the tolerance in Lemma 3.3 for rank six, choose $\gamma = \gamma_0/2$ and $C = 6/\gamma_0$; then $3/t_3 \le \gamma_0/2$. The loads and codegrees of the six-resource system satisfy the required bounds jointly.

Let m be the total number of selected supports and b_4 the number of marked ones. After undoing the pairs, the gain is $4m + b_4$: a pair yields two triangles, with gain four; a marked support yields gain five. Apply the first quota of (3.4) four times and the second once. The pair mass contributes at least $2t_3 - 6$, and the marked mass contributes $5t_4$. The losses sum to $5\epsilon|U| + 5D$, proving (3.6). Projection preserves the actual copies and their edge-disjointness.

C.1. Cleaning, selection, and realization

The proof starts with a regularity partition of $V(G)$. Contributions that do not admit a controlled transversal realization are discarded: irregular pairs, low-density pairs, and profiles with insufficient mass. A profile records the classes visited by a copy. Its type records whether the original copy is a triangle or a K_4.

The auxiliary selection object uses the edges of G as resources. Two copies sharing an edge compete for the same resource, even if their profile labels differ. Codegree bounds must therefore be verified for the joint family; rounding each profile separately is not enough. Forgetting the marks of a compatible selection produces a packing of actual copies.

The algebraic part can be isolated without losing this interpretation. Let S be the gain retained after cleaning and L its loss, so that $w(x) - L \le S$. If realization provides

$$(1 - u - v)S - g(\mathcal{P}) \le \zeta n^2,$$

with $u + v \ge 0$ and $S \le 5n^2/6$, then

$$w(x) - g(\mathcal{P}) \le L + \left(\zeta + \frac{5}{6}(u+v)\right) n^2. \tag{3.7}$$

To check this, write the difference as

$$w(x) - g(\mathcal{P}) = (w(x) - L - S) + ((1-u-v)S - g(\mathcal{P})) \\ + L + (u+v)S.$$

The first summand is nonpositive; the other three have exactly the budgets in (3.7). This separation prevents charging twice for the same cleaning step.

The realization with sufficient triangle mass and the small-mass case are combined before the final threshold is fixed. This last step introduces budgets of the form $15C$ and $13C + \zeta n^2$. The conditions

$$30C \le \xi n^2, \qquad 2\zeta \le \xi$$

bound both by ξn^2. The integer C and the selection constants are chosen uniformly, not from the reduced graph of a particular instance.

The selection step has a concrete description. Cleaning does not create abstract copies: for each active profile H it retains a fiber of actual copies in G. This fiber is assigned a volume $\mathrm{vol}(H)$ and a budget $(1+u)\,\mathrm{vol}(H)$. The retention lemmas prove

$$(1-v)\,\mathrm{vol}(H) \le |\,\mathrm{cleanFiber}(H)|. \tag{3.8}$$

Thus the mass entering the selector has already paid for the discarded patterns; they are not charged again at the end.

A hypergraph is then formed on the single resource set $E(G)$: each triangle contributes its three edges and each K_4 its six. Lemma 3.4 performs the selection after grouping triangle supports in pairs. The marks distinguish the K_4, and Lemma 3.3 simultaneously preserves the two quotas of the auxiliary system. Once the pairs have been undone, if there are a physical copies in total and b of them are K_4, the gain can also be written as

$$2(a-b) + 5b = 2a + 3b. \tag{3.9}$$

This identity agrees with $4m + b_4$ before undoing the pairs. The variables a and m count different objects; the unweighted nibble is not applied directly to the union of supports of sizes three and six.

It remains to verify (3.5) for the cleaned packing. The derivation below sums over profiles before applying Lemma 3.4. It therefore controls the shared physical resource, not merely the isolated contribution of each mark.

One explicit parameter hierarchy in the development takes

$$s = \min\{\xi/4500, 1/10\}, \qquad d = u = v = s, \qquad \delta = s^{21}/2208.$$

The bound on the number of regularity classes is then fixed, and finally a single N_ξ absorbs all remaining terms. The parameter ξ is not replaced by $1/n$.

The probabilistic proof of the nibble input is reused from Paper III [4]; Lemmas 3.3–3.4 explain the adaptation that turns its matching into the mixed packing of (3.1).

C.2. What is discarded and why the threshold is uniform

Three scales must be distinguished: the order n of the graph, the number k of classes in the regular partition, and the size t of each nonexceptional class. The number of classes satisfies $k_0 \le k \le B$, where B depends on the accuracy, not on the graph. A profile is the set of three

or four classes visited by a transversal copy. Write ψ_H for the fractional mass transferred to profile H, and

$$S = \sum_{H \text{ active}} g(H)\psi_H.$$

An "active" profile is one that has passed the density and mass filters; profiles are not normalized as if each had separate access to all edges.

The improved count distinguishes damaged copies from light profiles. Let N_3, N_4 be the numbers of possible profiles of each order. Then

$$w(x) - S \le L, \qquad L = 3\left(3\delta + \frac{1}{k_0} + d\right) n^2 + (N_4 + 2N_3)\theta. \tag{3.10}$$

Here δ is the regularity accuracy, d the density threshold and $\theta \ge 0$ the mass threshold per profile. The first count is made over discarded edges. A damaged triangle is removed, losing gain two. If a K_4 contains exactly one discarded edge, retain one of its triangles avoiding that edge: the loss is $5-2=3$. If it contains at least two, remove it entirely and charge $5 \le 3\cdot 2$. By the capacities, the sum of these charges is at most three times the number of discarded edges. Retained copies use subsets of the original resources, so their loads do not increase.

The second count uses a different operation. A K_4 of weight x in a light profile is replaced by its four triangular faces, each of weight $x/2$. Every edge lies in two faces: its load remains x, whereas the gain decreases from $5x$ to $4x$. The total cost is at most $N_4\theta$. Contributions to each triangle profile are then aggregated, and only after aggregation are light triangle profiles discarded, at cost at most $2N_3\theta$. Aggregating before discarding is essential: a face may receive mass from several profiles of order four.

The link to physical realization gives, for $\zeta > 0$ and $n \ge N_E(\zeta)$,

$$\begin{aligned} w(x) - g(\mathcal{P}) \le\ & \left(9\delta + \frac{3}{k_0} + 3d + \zeta + \frac{5}{6}(u+v)\right) n^2 \\ & + (N_4 + 2N_3)\theta. \end{aligned} \tag{3.11}$$

The assumptions are $\delta, d, \theta \ge 0$, $u + v \ge 0$, and $1 \le k_0 \le k$. The threshold N_E, defined in C.3, is chosen before the graph. The account does not leave the existence of a favorable packing as a hypothesis: the rounder is applied to the fractional packing reconstructed by the two operations above. It may use an auxiliary regular partition; we do not assert that every physical stage preserves a partition supplied in advance.

Thus the improvement in (3.10) is part of a complete loss bound, not merely an isolated saving on copies. The numerical schedule in C.3 remains conservative: reducing this coefficient does not by itself reduce the tower stated there.

We must still explain why physical rounding can be carried out simultaneously. The relevant codegree is the mass of copies containing **two given physical edges**, summed over all profiles and both types. Even if each profile behaves well separately, the same pair of resources could occur in several profiles. The estimate is therefore made before forgetting the marks. In the proof's notation, with

$$a_3 = d^3 - 3\delta > 0, \qquad a_4 = d^6 - 6\delta > 0,$$

joint control requires a scale of the form

$$k_3 a_4 + k_4 a_3 \le \gamma a_3 a_4 t, \tag{3.12}$$

where k_3, k_4 bound the numbers of active profiles of each type. Here is the count leading to that condition. Every active profile serves a pair of classes. Transferred copies consume edges of

that pair, whose capacities sum to at most t^2; hence $\psi_H \le t^2$. Replacing this bound by $\psi_H \le 1$ would be incorrect.

Each clean copy in the profile has weight ψ_H/b_H, where $b_H = (1+u)\,\mathrm{vol}(H)$. The counting estimates give $b_H \ge a_3 t^3$ for triangles and $b_H \ge a_4 t^4$ for K_4. Fix two distinct physical edges. In a triangle profile, at most one copy contains both. In a K_4 profile, there are at most t: if the edges share an endpoint they fix three vertices, leaving only the fourth to choose; otherwise they fix all four. Their respective contributions are therefore at most $1/(a_3 t)$ and $1/(a_4 t)$. Summing over all profiles gives

$$\sum_{K: e,f \in E(K)} x_K^{\mathrm{clean}} \le \frac{k_3}{a_3 t} + \frac{k_4}{a_4 t} = \frac{k_3 a_4 + k_4 a_3}{a_3 a_4 t} \le \gamma. \tag{3.13}$$

This proves the joint codegree condition in Lemma 3.4. Feasibility at a single physical edge needs a separate argument: a lower bound for the fibre volume does not by itself bound the load by one. The cleaning removes copies through roots with a large deviation on either side of their mean. Here a root is an edge in a prescribed pair of classes, not the clique root of the near construction.

Fix an active profile H, let $\mathcal{F}_H$ be its raw fibre, and write $V_H = |\mathcal{F}_H|$. For a pair q of its classes, let E_q be the actual edges in that pair and put $d_q = \max\{1, |E_q|\}$. If $c_{H,q}(e)$ counts the raw copies through $e \in E_q$, its reference value is $A_{H,q} = V_H/d_q$. This reference depends on the pair. Delete every copy using an edge for which $|c_{H,q}(e) - A_{H,q}| > uA_{H,q}$, for any of the three or six pairs in the profile.

An edge deleted in this test has no surviving copies through it. Every other edge has at most $(1+u)A_{H,q}$. Since a surviving copy receives weight $\psi_H/((1+u)V_H)$, this profile contributes at most ψ_H/d_q to the edge's load. The transferred capacity constraint is $\sum_{H \text{ serving } q} \psi_H \le d_q$; here the sum includes every active profile containing both classes of q, not one selected pair per profile. Summing the contributions therefore gives a total load at most one. The denominator is legitimate because an active fibre has positive volume.

It remains to check that this deletion is affordable. For one pair, the rooted second-moment estimates bound the number B_q of bad roots by $B_q u^2 a_3^2 \le 11\delta t^2$ for triangles and $B_q u^2 a_4^2 \le 23\delta t^2$ for four-cliques. A root supports at most t raw triangles or t^2 raw four-cliques. Summing over the three or six pairs bounds the numbers of deleted copies by $33\delta t^3/(u^2 a_3^2)$ and $138\delta t^4/(u^2 a_4^2)$, respectively. Multiple counting only enlarges these upper bounds. Comparing with the fibre lower bounds shows that the deleted fraction is at most v if $33\delta \le v u^2 a_3^3$ and $138\delta \le v u^2 a_4^3$.

The schedule in C.3 satisfies these inequalities: set $d = u = v = a$, use $\delta = a^{21}/2208$, and note that $a_3 \ge a^3/2$, $a_4 \ge a^6/2$ for $0 < a \le 1/10$. For example, $138\delta = a^{21}/16 \le a^{21}/8 \le v u^2 a_4^3$. Thus at least $(1-v)V_H$ copies remain. Their total weight is at least $(1-v)/(1+u) \ge 1-u-v$ times the transferred mass, as required in (3.8). Feasibility, retained mass and joint codegree have now been checked separately; none follows merely from the other two.

The marked theorem then supplies a packing whenever the triangle mass exceeds a constant C and

$$12 + 10D \le \zeta n^2, \tag{3.14}$$

where D is the selector's additive loss. If the triangle mass is below C, the small-mass branch replaces it using the triangle construction for small mass; its two losses are the quantities $15C$ and $13C + \zeta n^2$ cited above. Thus the branches cover all cases, and the final threshold absorbs C, D only once.

The order of choices matters as much as the inequalities. First fix the output accuracy. The selection lemma supplies its constants uniformly in the type of marks and in n. Next choose

δ, k_0, obtain the regularity bound B, and finally take N_ξ large enough for regularity, the class size required by (3.12), and the additive terms. This is the content of the uniform assembly. In particular, the number $4\cdot 10^{12}$ from the near-regime construction does not control this regularity threshold.

C.3. An explicit schedule and its propagation

For a rational accuracy $\xi > 0$, first fix

$$a = \min\{\xi/4500, 1/10\}, \quad \delta = a^{21}/2208, \quad z = \min\{\xi/10, 1\}, \quad k_0 = \lceil 1500/\xi \rceil + 1.$$

We use the following explicit regularity bound. For $\varepsilon > 0$, write $\lfloor\cdot\rfloor_+$ for the nonnegative integer floor and define

$$I(\varepsilon, k_0) = \max\left\{7, k_0, \left\lfloor \frac{\log(100/\varepsilon^5)}{\log 4} \right\rfloor_+ + 1\right\}, \qquad H(\varepsilon) = \lfloor 4/\varepsilon^5 \rfloor_+,$$
$$t_0 = I(\varepsilon, k_0), \qquad t_{j+1} = t_j 4^{t_j}, \qquad B(\varepsilon, k_0) = t_{H(\varepsilon)} 16^{t_{H(\varepsilon)}}.$$

In (C.1) we take $\varepsilon = \delta/8$ and $B = B(\delta/8, k_0)$. The marked selector, at accuracy z, supplies explicit constants $\gamma_* > 0, C_* \geq 0, D_* \geq 0$. Write

$$\gamma = \frac{1}{\lceil 1/\gamma_* \rceil}, \qquad C = \lceil C_* \rceil, \qquad a_3 = a^3 - 3\delta, \quad a_4 = a^6 - 6\delta.$$

The choice of δ ensures $a_3, a_4 > 0$. Ceilings in these expressions are nonnegative integers. One valid threshold is

$$N_E(\xi) = 1 + \max\Bigg\{k_0,\ \lceil B/\delta \rceil,\ \lceil (12 + 10D_*)/z \rceil,\ \left\lceil \frac{4B^5(a_3 + a_4)}{\gamma a_3 a_4} \right\rceil,\ \left\lceil \frac{50(1 + 2C)}{\xi} \right\rceil\Bigg\}. \tag{C.1}$$

The five terms have different roles: constructing the partition, controlling its exceptional part, paying the selector's additive loss, imposing joint codegree control and handling small triangle mass. None depends on G.

The connection to the cleaning in C.2 is worth making explicit. Apply that cleaning to the input packing, keeping both a value account and a triangle-mass account. If the latter is large, the two accounts allow the selector to be used with the constants in (C.1). If it is small, use the small-mass construction described after (3.14). The schedule dominates the losses in both branches and yields

$$w(x) - g(\mathcal{P}) \leq \xi n^2 \qquad (n \geq N_E(\xi)) \tag{C.2}$$

for every graph and every feasible fractional mixed packing. This is the numerical link that allows the improved account (3.11) to be used without retaining an unresolved realization hypothesis.

For Corollary 3.5, take $\xi = \eta/2$. Thus $N_{\mathrm{far}}(\eta) = N_E(\eta/2)$ suffices. At the accuracy used in the chordal assembly, $\eta_0 = 10^{-16}$, the parameters are

$$a = (9\cdot 10^{19})^{-1}, \qquad \delta = \frac{1}{2208(9\cdot 10^{19})^{21}}, \qquad z = (2\cdot 10^{17})^{-1}, \qquad k_0 = 3\cdot 10^{19} + 1.$$

The three integer selector quantities to be absorbed are

$$G_1 = \lceil 1/\gamma_* \rceil, \quad G_2 = \lceil C_* \rceil, \quad G_3 = \lceil 2\cdot 10^{17}(12 + 10D_*) \rceil.$$

To see how they are absorbed, fix $\beta_0 = (64 \cdot 10^{18})^{-1}$. We give the schedule explicitly, so that the constants in Table C.1 have specified meanings. For $r \geq 2$ and $0 < \beta \leq 1/2$, all the quantities below are evaluated at (r, β), unless another argument is displayed:

$$a_T = (r-1)/r, \quad L_T = -\log \beta, \quad M_T = 16rL_T + 1,$$
$$\gamma_T = \min\{1/8, e^{-8M_T - 1}/32\}, \quad \epsilon_T = 4a_T\gamma_T, \quad q_T = 1 - a_T\gamma_T, \quad T_T = \lceil M_T/\gamma_T \rceil,$$
$$G_T = (2 + r/(\epsilon_T\gamma_T))^2, \quad \eta_T = \frac{\beta}{8(2G_T)^{T_T}}, \quad \ell_T = \frac{3}{4}q_T^{T_T}.$$

The round count is T_T; η_T bounds the exceptional fraction, and ℓ_T is the retained degree scale. To pass to the selector, set

$$D_{0,T} = \frac{256r}{(\beta/2)^2\gamma_T} + \frac{96}{\epsilon_T} + 4, \qquad c_{0,T} = \frac{\eta_T\epsilon_T^2\gamma_T(\beta/2)^2}{16384r},$$
$$\mu_T = \min\{8\gamma_T, c_{0,T}\ell_T, [2(D_{0,T}+1)]^{-1}, 1/2\}, \qquad d_{0,T} = \max\{1, D_{0,T}/\ell_T\},$$
$$D_S(r,\beta) = 1 + \max\{\lceil d_{0,T} \rceil, \lceil 4(1+r^2)/\mu_T \rceil\},$$
$$\gamma_R(r-1,\beta) = \min\{1/D_S(r,\beta), \mu_T/4, 1\}.$$

For general positive accuracy, the selector construction uses $\min\{\beta, 1/2\}$; this equals β in the range just stated. The rank increases from six to seven when the selector is converted to the nibble input. Substituting $r = 7$ and $\beta = \beta_0$ gives the inequalities recorded below. They are bounds proved for this schedule, not decimal approximations or free choices of the selector constants.

Step	Verified bounds
Schedule	$L_T(\beta_0) \leq 46, \quad M_T(7,\beta_0) \leq 5153$
Intermediate chain	$\gamma_T(7,\beta_0) \geq 2^{-59485}, \quad T_T(7,\beta_0) \leq 2^{59498}$
Selector	$D_S(7,\beta_0) \leq 2^U, \quad \gamma_R(6,\beta_0) \geq 2^{-U}$
Three final integers	$G_1 \leq 2^{U+2}, \quad G_2 \leq 2^{U+3}, \quad G_3 \leq 2^{U+67}$

Table C.1. Numerical absorption of the selector. Here $U = 2^{59516} + 191432$.

Here is the numerical chain behind the table. Since $\beta_0 \geq 2^{-66}$, we have $L_T \leq 66 \log 2 \leq 46$ and $M_T \leq 5153$. The inequality $8 \cdot 5153 + 1 \leq 59480 \log 2$ gives $\gamma_T \geq 2^{-59485}$. Put $a_0 = 59485$ and $Y = 2^{a_0+31}$; these are numerical auxiliaries, not graph parameters. Then $\epsilon_T = (24/7)\gamma_T \geq 2^{-a_0}$ and $T_T \leq 5153\, 2^{a_0} + 1 \leq 2^{a_0+13}$. Moreover, $2G_T \leq 2^{4a_0+5}$ and $4a_0 + 5 \leq 2^{18}$, so $(2G_T)^{T_T} \leq 2^Y$ and $\eta_T \geq 2^{-(Y+69)}$.

For the retained scale, write $x = 6\gamma_T/7 \leq 1/2$. The inequalities $1 - x \geq (1+2x)^{-1}$ and $\gamma_T T_T \leq M_T + \gamma_T \leq 5153 + 1/8$ give $2xT_T \leq (12/7)(5153 + 1/8) \leq 12748 \log 2$. Consequently $q_T^{T_T} \geq 2^{-12748}$ and $\ell_T \geq 2^{-12749}$. Substitution in the definitions gives $D_{0,T} \leq 2^{a_0+146}$ and $c_{0,T} \geq 2^{-(Y+3a_0+220)}$. Let $Z = 3a_0 + 220 + 12749 = 191424$. Each of the four entries defining μ_T is at least $2^{-(Y+Z)}$. Also $d_{0,T} \leq 2^{a_0+146+12749}$. The ceiling terms in D_S are therefore bounded by $256\, 2^{Y+Z} = 2^U$, where $U = Y + 191432$; the extra unit in D_S is included in this bound. Both $1/D_S$ and $\mu_T/4$ are at least 2^{-U}, which proves the selector row.

Finally, substituting this lower bound into the three selector quantities gives the last row. For its largest entry, the intermediate denominator is at most $10\, 2^U + 28$; hence its ceiling is at most $2 \cdot 10^{17} \cdot 392\, 2^U \leq 2^{U+67}$. Since $U + 67 \leq 2^{59517}$, the four rows imply $G_i \leq 2^{\,2^{59517}}$. We now give the remaining comparison.

The regularity iterate at the chosen accuracy. At $\eta_0 = 10^{-16}$, put $\varepsilon = \delta/8$. The exact inequality $100/\varepsilon^5 \leq 4^{4000}$ implies that the floor in $I(\varepsilon, k_0)$ is at most 4000. Since $k_0 = 3 \cdot 10^{19} + 1$,

the maximum defining I is therefore k_0. Also,

$$H(\varepsilon) = 4 \cdot 8^5 \cdot 2208^5 (9 \cdot 10^{19})^{105} = 4 \cdot 8^5 \cdot 2208^5 \cdot 4500^{105} (2 \cdot 10^{16})^{105} = h.$$

Thus the iteration count is exactly the integer in §6.3. Write $R = t_h$, so $B = R16^R$. For every integer $x \geq 1$, $x4^x \geq 2^x$. The iterates are nondecreasing, and hence

$$R \geq t_2 \geq 2^{2^{k_0}} \geq 2^{2^{59517}} \geq G_i \qquad (i = 1, 2, 3).$$

In particular, $R \geq 2^{2000}$. The two-iteration comparison is a lower bound for the regularity iterate, not an upper bound on the number of iterations needed by regularity.

Absorbing the five terms. At these parameters, (C.1) and the definitions of G_i give

$$N_{\mathrm{far}}(\eta_0) \leq k_0 + \alpha B + G_3 + \beta G_1 B^5 + 10^{18}(1 + 2G_2) + 1,$$
$$\alpha = 2208(9 \cdot 10^{19})^{21}, \qquad \beta = 8(9 \cdot 10^{19})^6.$$

Here the maximum in (C.1) is bounded by the sum. The fourth term uses $a_3 \geq a^3/2$, $a_4 \geq a^6/2$, and the sharper direct substitution $(a_3 + a_4)/(a_3 a_4) \leq 2(9 \cdot 10^{19})^6$; all coefficients are rational. Each of $k_0, \alpha, \beta, 10^{18}$ is at most 2^{2000}, and therefore at most R. Since $16^R \geq 4R$, we have $B \geq 4R^2$ and $B \geq 4$. It follows that

$$\beta G_1 B^5 \leq R^2 B^5 \leq B^6/4.$$

The other terms, including the final unit, sum to at most $7B^2$: the bounds for $k_0, \alpha B, G_3, 10^{18}(1+2G_2), 1$ are respectively $B^2, B^2, B^2, 3B^2, B^2$. Moreover, $B \geq 4$ gives $256B^2 \leq B^6$. Consequently

$$N_{\mathrm{far}}(\eta_0) \leq B^6/4 + 7B^2 \leq (1/4 + 7/256)B^6 \leq B^6.$$

Converting the iterate to a tower. Recall $T(0) = 1$, $T(j+1) = 2^{T(j)}$. The inequality $4x \leq 4^x$ for integers $x \geq 1$ gives $4(x4^x) \leq 2^{4x}$. Starting from $4k_0 \leq 2^{67} \leq T(5)$, induction on j yields $4t_j \leq T(j+5)$. Put $q = T(h+5)$; then $4R \leq q$ and $q \geq 6$. Finally, $R \leq 2^R$ and $8q \leq 2^q$ imply

$$B^6 = (R16^R)^6 \leq 2^{30R} \leq 2^{2^q} = T(h+7),$$

where $30R \leq 8q \leq 2^q$. Combining the absorption and this tower comparison proves

$$N_{\mathrm{far}}(\eta_0) \leq T(h+7). \tag{C.3}$$

The comparison is symbolic: it does not require constructing an integer with that many digits. Nor does it assume a result on resolvable designs or a small-separator regime. Hence (C.3) holds for the far branch of **every graph**; chordality enters when this branch is combined with the near branch in §6.

The threshold also admits a bound uniform in the margin. For every rational $0 < \eta \leq 1$, the same schedule satisfies

$$N_{\mathrm{far}}(\eta) \leq T\left(\left\lceil \frac{c_{\mathrm{poly}}}{\eta^{105}} \right\rceil\right), \qquad c_{\mathrm{poly}} = 4 \cdot 17664^5 \cdot 9000^{105} + 3006. \tag{C.4}$$

It bounds the selector, regularity and cleanup parameters before the graph is chosen; η is not allowed to depend on the input graph after the threshold has been fixed. The specialized numerical bound in §6.3 remains valid. Formula (C.4) is the additional comparison needed for the growing-defect exact band in Corollary 6.6.

Appendix D. Complete graphs and comparator identities

D.1. The complete graph at every order

The target without an additive term can be verified on an infinite family without using the eventual threshold. This construction is not used in the proof of Theorems A and B.

Proposition D.1. For every integer $n \geq 0$, there is a partition of K_n into pieces of order at most four with at most $M(n)$ pieces.

Proof. For $n = 0, 1$, use the empty partition. For $n \geq 2$, it suffices to construct a mixed packing $\mathcal{P}$ with

$$n(n-2) \leq 3g(\mathcal{P}). \tag{D.1}$$

Indeed, its completion satisfies

$$|Q| = \binom{n}{2} - g(\mathcal{P}) \leq \frac{n(n-1)}{2} - \frac{n(n-2)}{3} = \frac{n(n+1)}{6}. \tag{D.2}$$

Since $|Q|$ is an integer, this gives the floor $M(n)$.

The construction uses two block frameworks on $(\mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}) \sqcup \{\infty\}$. For odd M it uses the Bose framework [19]; for even M, the Skolem framework [20]. Each covered edge belongs to a unique block, and the blocks are triangles or K_4. In the odd framework, triangle columns can be replaced by K_4 columns containing ∞. The six residue classes are assembled as follows; the names refer to the implemented constructions, not to a new attribution of the classical designs.

Order n	Blocks used	Final step
$3M$, M odd	Triangle Bose construction	Covers K_n
$3M-1$, M odd	Triangle Bose construction on $n+1$ vertices	Delete one vertex and its blocks
$3M+1$, M even, $M \geq 2$	Triangle Skolem construction	Covers K_n
$3M$, M even, $M \geq 2$	Triangle Skolem construction on $n+1$ vertices	Delete one vertex and its blocks
$3M+1$, M odd	Bose construction with M columns of K_4	Covers K_n
$3M+2$, M odd	Construction from the preceding row	Add one vertex; complete its edges as K_2

Table 6. Constructions for the complete case. In the last row the added vertex does not participate in the packing; it is not an isolated vertex of the complete graph.

Let us check the gain required by the assembly. For disjoint mixed blocks, if E_{covered} is their covered edge set and b_4 their number of K_4, the contributions two and five give

$$3g(\mathcal{P}) = 2|E_{\text{covered}}| + 3b_4. \tag{D.3}$$

The rows covering K_n satisfy (D.1), since $2|E_{\text{covered}}| = n(n-1) \geq n(n-2)$. If one vertex is deleted from a triangle decomposition of K_{n+1}, it belongs to $n/2$ triangles: its n incident edges are grouped in pairs. There remain $n(n+1)/6 - n/2 = n(n-2)/6$ triangles, whose gain satisfies (D.1) with equality. Finally, the last row covers the edges of K_{3M+1} with M blocks of K_4, so that

$$3g(\mathcal{P}) = (3M+1)3M + 3M = (3M+2)3M = n(n-2). \tag{D.4}$$

The six rows exhaust the residue classes modulo six for $n \geq 2$. Combining these counts and applying completion (D.2) proves the claim.

This proves $b = 0$ for complete graphs, not for all small chordal graphs. In one congruence class, pieces of order four are necessary.

Proposition D.2. If $n \equiv 4 \pmod 6$, every partition of K_n into pieces of order at most three has more than $M(n)$ pieces.

Proof. Let a be the number of K_2 pieces and t the number of triangles. Counting edges and pieces gives

$$a + 3t = \binom{n}{2}, \qquad |Q| = a + t = \frac{\binom{n}{2} + 2a}{3}. \tag{D.5}$$

Each vertex has odd degree $n-1$. Incident triangles contribute in pairs, so the vertex must belong to at least one K_2 piece. Hence $2a \geq n$, and

$$|Q| \geq \frac{\binom{n}{2} + n}{3} = \frac{n(n+1)}{6}.$$

For $n = 6r+4$, the last quantity has fractional part $1/3$. Since $|Q|$ is an integer, $|Q| \geq M(n)+1$.

For example, K_{10} requires at least 19 pieces if only edges and triangles are allowed, whereas Proposition D.1 gives at most $M(10) = 18$ when K_4 is also allowed. The order-four restriction in the sharp bound cannot uniformly be replaced by order three. This obstruction is not extended to residue five.

D.2. Equality of mixed optima in complete-split graphs

Proposition D.3 (zero mixed gap in complete-split graphs). Let $S = K_k \vee I_h$, with $2 \leq k \leq h$, and let $W(S)$ be the maximum gain of a physical mixed packing. Then

$$W^*(S) = W(S) = 2\binom{k}{2}. \tag{D.6}$$

To justify the fractional bound, let $j(K)$ count core edges contained in a mixed piece K. A clique of S contains at most one host. A triangle with a host has $j(K) = 1$ and gain two; a K_4 with a host has $j(K) = 3$ and gain five. Pieces lying entirely in the core also satisfy $g(K) \leq 2j(K)$. Thus, for every feasible fractional packing x,

$$\begin{aligned} w(x) &\leq 2\sum_K j(K)x_K \\ &= 2\sum_{e \in E(K_k)} \sum_{K : e \in E(K)} x_K \leq 2\binom{k}{2}. \end{aligned} \tag{D.7}$$

The last inequality uses capacity one on each core edge. The construction in Section 6.1 contains one triangle per internal base and attains this gain. Since every integral packing defines a fractional one, both equalities in (D.6) follow.

The conclusion is an equality of optimal values in this family under $2 \leq k \leq h$. It does not assert that every vertex of the fractional polytope is integral, nor does it extend zero gap to arbitrary split graphs.

D.3. Exact reserve from core displacement

Identity (6.3) also retains the floor. For integers $0 \leq k \leq n$ with $\binom{k}{2} \leq k(n-k)$, define

$$d_{n,k} = \begin{cases} n - 3k, & 3k \leq n, \\ 3k - n - 1, & 3k > n. \end{cases}$$

The reserve of the complete-split profile is then exactly

$$M(n) = B_n(k) + M(d_{n,k}). \tag{D.8}$$

Indeed, in both branches $d_{n,k}(d_{n,k}+1) = (n-3k)(n-3k+1)$. Substituting into (6.3), dividing by six and taking floors, the integer $B_n(k)$ comes outside the floor, yielding (D.8). The hypothesis on $\binom{k}{2}$ makes $B_n(k)$ nonnegative.

This form identifies the available margin without replacing M by its rational envelope. In particular, the reserve is zero exactly when $d_{n,k} \le 1$, recovering the optimal cores in (6.4). It does not by itself supply a partition of an arbitrary chordal graph: it quantifies the profile budget against which the construction is compared.

Appendix E. The fixed-defect extension

Theorem C is not obtained by replacing "chordal" with "defect s" in every stability lemma. Its proof uses a minimum-degree terminal, a constructor on actual resources and an argument removing an additive constant.

E.1. Constructor and accounting

Localization and normalization give $V = S \sqcup H \sqcup W$: S is the base region, H the host region and W the exceptional region. Not every pair in S is required to be an edge. The constructor hosts only actual edges of $G[S]$.

First absorb the vertices $w \in W$. For each, choose a matching M_w of links ah between S and H, with both endpoints adjacent to w; each link gives the triangle wah. Balanced selection controls spoke consumption by row and column. Subsequent phases exclude occupied spokes.

In the first phase, color the edges of $G[H]$ with k colors and balance the classes to have size at most q. Averaging cyclic shifts of their assignment to hosts in S gives t_I hosted bases with $2t_I \ge e(G[H])$. In the second phase, list hosting realizes all edges of $G[S]$ using the remaining spokes. The dyadic construction applies Galvin to auxiliary bipartite problems, not to an arbitrary graph.

Among the conditions checked before the phases are

$$\begin{gathered} d_{\max}(G[H]) + 1 \le k, \quad |S| \le k \le 2(|S| - 2\sigma), \quad q \ge \lceil e(G[H])/k \rceil, \\ 3\lceil |S|/2 \rceil + 2\tau + 4q \le |H| + 2. \end{gathered} \tag{E.1}$$

Here $d_{\max}$ denotes maximum degree. The parameters σ, τ bound row and column absences, including reserves for W; they are not chosen after constructing the partition. The three triangle families are edge-disjoint. Their completion satisfies

$$|Q| + 2\left(\sum_{w \in W} |M_w| + t_I + e(G[S])\right) = e(G). \tag{E.2}$$

We now expand this account. Write $c = |S|$, $b = |H|$ and $t_W = |W|$, so that $n = c + b + t_W$. For each $w \in W$, let $u_w = |N(w) \cap S|$, $v_w = |N(w) \cap H|$, and put

$$d_W = \sum_{w \in W} \deg(w), \qquad m_W = \sum_{w \in W} \min(u_w, v_w).$$

Let D bound the number of missing links between S and H, and choose an integer $L \ge \lfloor \sqrt{D} \rfloor$. Absorption gives $\min(u_w, v_w) \le |M_w| + t_W + L$. Also, $e(G) \le e(G[S]) + e(G[H]) + cb + d_W$: edges inside W are counted twice in the last summand, which is permissible in this upper bound. Substitution into (E.2), using $2t_I \ge e(G[H])$ and summing the matching estimates, yields

$$|Q| + e(G[S]) + 2m_W \le cb + d_W + 2t_W(L + t_W). \tag{E.2a}$$

The role of rooted defect is to pay the right-hand side of this account. The joint inequality derived from the normalized data is

$$\begin{aligned} d_W + 2t_W(L + t_W) + \binom{c}{2} + \binom{s+1}{2} \\ \le e(G[S]) + 2m_W + sc + \frac{t_W(c+b+s)}{3}. \end{aligned} \tag{E.2b}$$

Adding (E.2a) and (E.2b) cancels d_W, $2t_W(L + t_W)$, $e(G[S])$ and $2m_W$. For the same physical partition Q, this leaves

$$|Q| + \binom{c}{2} + \binom{s+1}{2} \le c(b+s) + \frac{t_W(c+b+s)}{3}. \tag{E.2c}$$

No second partition is chosen to satisfy this last inequality.

We explain the origin of (E.2b), since it does not follow from the physical accounting alone. Normalization starts from a clique A of size close to $n/3$. Its vertices with many missing links are removed; outside it, vertices of large exterior degree are separated from those of small exterior degree. Among the former, those with few missing links to the light region are added to S, and those with few missing links to the retained root are added to H. The remaining vertices form W. Let $\mathcal{L} \subseteq H$ denote the light region. The scales are

$$\rho = \frac{n}{(s+1)^2}, \qquad \lambda = \frac{n}{(s+1)^4}.$$

For $n \ge 10^{50}(s+1)^8$, normalization and the choice of reserves give

$$t_W \le \frac{4\rho}{10^{27}}, \qquad L \le \frac{\rho}{10^{13}} + 1, \qquad |\mathcal{L}| - \frac{n}{3} \le c + \frac{4\lambda}{10^{31}}.$$

Here are the counts behind these estimates. Put $h = s+1$ and $\varepsilon = 1/(10^{41}h^8)$. The input to normalization is a clique A with $\big||A| - n/3\big| \le \varepsilon n$, at most εn^2 missing links to its complement and at most εn^2 exterior edges, together with minimum degree at least $(1/3 - \varepsilon)n$. These are the localized inputs, not conclusions of rooted defect alone. Notice that $\varepsilon n^2 = \lambda^2/10^{41}$ and $n\lambda = \rho^2$.

Remove from A every vertex missing more than $\lambda/10^{10}$ exterior neighbors, and call the retained set A_0. Counting the missing links by their endpoint in A gives $|A \setminus A_0| \le \lambda/10^{31}$. Hence $n/3 - 2\lambda/10^{31} \le |A_0| \le n/3 + \lambda/10^{41}$. Moving a vertex to the exterior creates at most n additional exterior edges, so the new exterior $R_0 = V \setminus A_0$ has at most $2\rho^2/10^{31}$ edges. Its heavy set Y, defined by degree at least $\rho/10^4$ within R_0, therefore has size at most $4\rho/10^{27}$, by summing degrees. The light set is $\mathcal{L} = R_0 \setminus Y$.

Partition Y as follows, in the indicated order. The set Y_c consists of vertices missing at most $\rho/40$ neighbors in $\mathcal{L}$. Of the remaining vertices, put in Y_r those missing at most $\rho/200$ neighbors in A_0. The rest form W. Thus $S = A_0 \cup Y_c$ and $H = \mathcal{L} \cup Y_r$. In particular, $t_W \le |Y|$, $|c - n/3| \le \rho/10^{20}$, and $m = e(G[H]) \le 2\rho^2/10^{31}$. Also $|\mathcal{L}| \le 2n/3 + 2\lambda/10^{31}$ and $c \ge n/3 - 2\lambda/10^{31}$, giving the displayed bound on $|\mathcal{L}| - n/3$.

For a light vertex, minimum degree and the bound on its exterior degree imply at most $\rho/10^4 + 2\lambda/10^{41} < \rho/200$ missing neighbors in A_0. The definition gives the same $\rho/200$ bound for Y_r. Adding Y_c and the $2t_W$ reserved spokes therefore gives $\sigma \le \rho/200 + 12\rho/10^{27}$. On the other side, every vertex of S misses at most $\rho/40$ light neighbors; adding Y_r and the reserves gives $\tau \le \rho/40 + 12\rho/10^{27}$. These are the row and column bounds used again in E.4.

Finally, let D count all missing links between S and H. Splitting them between $A_0 : \mathcal{L}$, $A_0 : Y_r$ and $Y_c : H$ gives $D \le \lambda^2/10^{41} + |Y_r|\rho/200 + |Y_c|(\rho/40 + |Y_r|) \le (\rho/10^{13})^2$. Taking $L = \lfloor\sqrt{D}\rfloor + 1$

yields $D < L^2$ and $L \le \rho/10^{13} + 1$. These estimates precede the execution of the constructor. In particular, the reserves are not adjusted retrospectively to the piece count.

Let ν be the maximum number of disjoint nonedge pairs within S. The rooted-defect condition, together with the normalized common-neighbor bounds, implies $\nu \le s$; the argument is given in E.4 below. This is a property of the normalized core, not of every vertex subset in a graph of rooted defect at most s. The extremal matching bound applied to the complement of $G[S]$ gives

$$M_S := \binom{c}{2} - e(G[S]) \le \nu c - \binom{\nu+1}{2}.$$

To measure each exception's contribution, define

$$g_w = \deg(w) - 2\min(u_w, v_w) + 2(L + t_W) - \frac{c+b+s}{3},$$
$$e_1 = 4t_W + 2L + \frac{4\rho}{10^{27}}.$$

Thus (E.2b) is equivalent to $\sum_{w\in W} g_w + M_S + \binom{s+1}{2} \le sc$. For the stability argument, retain the residual credit

$$R_* = E - \sum_{w\in W} g_w, \qquad E = e(G[S]) - \binom{c-s}{2} = e(G[S]) - \binom{c}{2} - \binom{s+1}{2} + sc.$$

Equation (E.2b) asserts $R_* \ge 0$; E.4 will retain a strictly positive allowance per exceptional vertex. Throughout Appendix E, $t_W = |W|$ counts exceptions, t_I counts first-phase triangles, and r is reserved for the additional vertices moved out of the core when extracting an actual clique. The exceptions that may contribute positively lie in

$$X = \{w \in W : v_w - u_w > n/3 - \rho/400\}, \qquad j = |X|.$$

Outside X, $g_w \le 0$; inside it, $g_w \le |N(w) \cap \mathcal{L}| - n/3 + e_1$.

The defect condition also supplies a joint bound, rather than only vertex-by-vertex estimates:

$$\sum_{w\in X'} |N(w) \cap \mathcal{L}| \le (s-\nu)|\mathcal{L}| + \frac{n}{9} \quad \text{if } X' \subseteq W, \quad |X'| \le 2(s+1). \tag{E.2d}$$

To derive (E.2d), write $x = |X'| \le 2h$ and fix ν disjoint nonedge pairs in S, with endpoint set B. Each $w \in X' \subseteq W$ misses more than $\rho/200$ neighbors in A_0. Since $\rho \ge 10^{50}h^2$, we may greedily choose distinct nonneighbors $a_w \in A_0 \setminus B$. Let $U_0 = B \cup \{a_w : w \in X'\}$, and let $Z \subseteq \mathcal{L}$ consist of the common neighbors of U_0. Every clique in $\mathcal{L}$ has size at most $\omega = \lceil \rho/10^4 \rceil$, by the light exterior-degree bound.

Apply rooted defect with empty root to successive induced subgraphs of $U_0 \cup X' \cup Z$, retaining U_0. There are three possibilities for the vertex supplied by that condition. If it is $z \in Z$, its neighborhood contains the ν fixed missing pairs and, for every remaining adjacent exception w, the missing pair $\{w, a_w\}$. These pairs are disjoint. A clique omits at least one endpoint of each, so at most $s - \nu$ such exceptions can remain adjacent to z. Delete this row and charge at most $s - \nu$ incidences. If the selected vertex is an exception w, it has at most $\omega + s$ neighbors in the remaining light set; delete it and charge those incidences. If the selected vertex lies in U_0, it is adjacent to every remaining vertex of Z. There are then at most $\omega + s$ remaining rows, and we stop, charging at most $(\omega + s)x$ incidences. This procedure either deletes a vertex or stops, and therefore terminates. The three charges give $e(Z, X') \le (s-\nu)|Z| + 2(\omega + s)x$.

Restore the omitted light vertices. Each endpoint in B misses at most $\rho/40$ of them, and each $a_w \in A_0$ at most $\lambda/10^{10}$. Thus $|\mathcal{L} \setminus Z| \le 2\nu\rho/40 + x\lambda/10^{10}$. Each restored row contributes at most x incidences, so

$$e(\mathcal{L}, X') \le (s-\nu)|\mathcal{L}| + 2(\omega+s)x + x\left(\frac{2\nu\rho}{40} + \frac{x\lambda}{10^{10}}\right).$$

To obtain the error $n/9$ in (E.2d), use $\nu \le s < h$, $x \le 2h$, $h^2\rho = n$, $h^2\lambda = \rho$ and $h^2 \le \rho/10^{50}$. The error just displayed is at most $n/10 + 4n/10^4 + 4\rho/10^{10} + 4\rho/10^{50} \le n/9$. Applying it to $2s+2$ large exceptions also gives $j \le 2s+1$.

The budget proof now has three cases.

1. If $j = 0$, every contribution g_w is nonpositive. It suffices to use

$$sc - \nu c - \binom{s+1}{2} + \binom{\nu+1}{2} = (s-\nu)\left(c - \frac{s+\nu+1}{2}\right) \ge 0,$$

   where normalization gives $c \ge s+\nu+1$.
2. If $1 \le j \le s-\nu$, each exception in X misses more than $\rho/40$ light neighbors. After paying the error terms, this gives $g_w \le c - \rho/80$. Hence

$$\sum_{w\in W} g_w + M_S + \binom{s+1}{2} \le (j+\nu)c - \frac{j\rho}{80} - \binom{\nu+1}{2} + \binom{s+1}{2} \le sc.$$

   The last inequality uses $j+\nu \le s$ and $\binom{s+1}{2} \le \rho/(2\cdot 10^{50})$.
3. If $j \ge s-\nu+1$, apply (E.2d) to X. Since $n/3 - e_1 \ge 0$, we may subtract at least $(s-\nu+1)(n/3-e_1)$. Using the bound on $|\mathcal{L}|$ then gives

$$\sum_{w\in W} g_w \le (s-\nu)c - \frac{2n}{9} + \frac{4(s+1)\lambda}{10^{31}} + (s+1)e_1.$$

   The negative term pays the remaining errors and $\binom{s+1}{2}$: substitute $e_1 \le 20\rho/10^{27} + 2\rho/10^{13} + 2$, $n = \rho(s+1)^2$ and $\rho \ge 10^{50}(s+1)^2$. Together with the bound on M_S, this again proves (E.2b).

Combining these three cases gives the joint inequality. They do not assert that absorption, without the rooted-defect condition, always has a favorable balance.

Finally, put $z = \lceil (c+b+s)/3 \rceil$. The integer comparator's parabola and the sum of its increments give, when $c+b \ge s+2$,

$$\begin{aligned} c(b+s) &\le \binom{c}{2} + M(c+b+s), \\ Q_s(c+b) + t_W z &\le Q_s(c+b+t_W). \end{aligned} \tag{E.2e}$$

Inserting the first inequality into (E.2c) gives $|Q| \le Q_s(c+b) + t_W z$; the second yields $|Q| \le Q_s(n)$. The construction and this arithmetic step give both (E.2c) and the target bound for the same partition.

The list bound and first-phase capacity window have the same form as conditions in [15, Lemma 4.1 and (4.4)]. Here they are justified by dyadic hosting, averaging over cyclic shifts, and balanced absorption. This comparison identifies the mechanisms and their shared conditions; it claims neither priority for those conditions nor historical independence.

E.2. From the terminal to the eventual theorem

The selected explicit proof follows the terminal and minimum-degree reduction described below. Its localization is supplied by E34, and its thresholds are evaluated explicitly; the historical existential assembly has the same reduction structure.

For each s, the terminal result supplies $\epsilon_s > 0$ and a threshold beyond which every graph in the class with $\delta(G) \geq (1/3 - \epsilon_s)n$ admits a partition of cost $Q_s(n)$. If there is a quadratic fractional margin, Corollary 3.5 applies to the graph, and $M(n) \leq Q_s(n)$ in the range considered. Otherwise, fixed-defect localization gives the normalized data of E.1. The local threshold $\lceil 10^{50}(s+1)^8 \rceil$ for that terminal is not the global N_s, which includes localization and rounding.

The resulting threshold is $N_s = F_s = \mathrm{Fexp}(\mathrm{NeditE}, s)$. Its chordal-edit step follows [22], with the adaptations in Appendix F.5. Corollary 6.6 bounds F_s by an explicit tower.

Here is why a minimum-degree terminal suffices. For sufficiently large n,

$$Q_s(n) - Q_s(n-1) = \left\lfloor \frac{n+s+1}{3} \right\rfloor. \tag{E.3}$$

The class is hereditary under induced subgraphs. A first induction gives $c_4(G) \leq Q_s(n) + K_s$, with constant K_s: small orders are paid for by edges; if a vertex has degree at most (E.3), delete it and restore its incident edges; otherwise, apply the terminal.

Repeating that deletion does not suffice to remove K_s. Now take n larger still, with $\epsilon_s n \geq K_s + 1$. If a vertex satisfies

$$\deg(v) + K_s \leq \left\lfloor \frac{n+s+1}{3} \right\rfloor, \tag{E.4}$$

the established additive bound for $G - v$, together with (E.3), gives $c_4(G) \leq Q_s(n)$ directly. If none satisfies (E.4), every vertex has degree at least $(1/3 - \epsilon_s)n$, and the terminal applies. This removes the constant without assuming the sharp bound at small orders.

E.3. An unrestricted lower witness

Suppose $n \geq 2s+2$ and set $k = \lfloor (n+s+1)/3 \rfloor$. Take disjoint sets C, D, H of sizes $k-s, s, n-k$. Make C complete and D, H independent, and add all links between $C \cup D$ and H, with none between C and D.

This graph has rooted defect at most s. In any induced subgraph with a prescribed root, a vertex of H outside the root has a clique of neighbors in C and at most s neighbors in D. If all vertices of H in the induced subgraph lie in the root, there is at most one, since H is independent; any remaining vertex of C or D is then simplicial.

Assign weight -1 to edges within C and $+1$ to links towards H. Every clique has weight at most one: if it contains a vertex of D, it is at most an edge towards H; if it contains r vertices of C and one of H, it has weight $r - \binom{r}{2} \leq 1$; cliques within C have nonpositive weight. Thus every partition, with no restriction on the order of its pieces, needs at least

$$k(n-k) - \binom{k-s}{2} = M(n+s) - \binom{s+1}{2} = Q_s(n) \tag{E.5}$$

pieces. The equality is the integer comparator identity for this choice of k. Together with the eventual upper bound, it proves the maximum in Theorem C.

E.4. Retained accounts for same-root stability

We derive (6.14) under the normalized terminal hypotheses of E.1. Write $c = |S|$, $b = |H|$, $t_W = |W|$, $n = c + b + t_W$, and retain the scales $\rho = n/(s+1)^2$ and $\lambda = n/(s+1)^4$. Let $a = cb - e(S,H)$ be the number of missing cross-links and $m = e(G[H])$. The exceptional contributions g_w, the core excess E, and the residual credit R_* are those defined in E.1. The argument uses the normalized terminal, including its minimum-degree, missing-link and exterior-degree bounds; rooted defect alone does not supply these bounds.

A smaller palette leaves a measurable saving. Set

$$d = \lfloor \rho/1000 \rfloor, \qquad \kappa = 2(c - 2\sigma) - d, \qquad q = \lceil m/\kappa \rceil, \qquad \theta_s = \frac{1}{2000(s+1)^2}. \tag{E.6}$$

Here σ, τ are chosen before the construction, as the largest missing-link counts in the respective rows and columns, with $2t_W$ added for reserved spokes. Normalization gives

$$|c - n/3| \le \rho/10^{20}, \qquad |b - 2n/3| \le \rho/10^{20},$$
$$\sigma \le \rho/200 + 12\rho/10^{27}, \qquad \tau \le \rho/40 + 12\rho/10^{27}, \qquad m \le 2\rho^2/10^{31}.$$

The terminal threshold gives $\rho \ge 10^{50}$, so $\rho/2000 \le d \le \rho/1000$. Substitution in (E.6) yields $c \le \kappa \le n$ and $\kappa \ge n/2$. More explicitly, $\kappa \ge 2n/3 - 11\rho/500$. A light host has exterior degree at most $\rho/10^4$; each of the remaining hosts misses more than $\rho/40$ vertices of the light region, so its degree within H is at most $b - \rho/40$. Both degree bounds, with one added for coloring, are at most κ. For the second comparison, the gap between $\rho/40$ and $11\rho/500$ pays $\rho/10^{20} + 1$.

Since $\kappa \ge n/2$ and $\rho \le n$, the edge bound gives $q \le 4\rho/10^{31} + 1$. The same displayed estimates give $3\lceil c/2 \rceil + 2\tau + 4q \le b + 2$. Thus all the capacity and list conditions in (E.1) still hold with this smaller palette. No new coloring assumption is introduced.

The cyclic averaging step now retains t_I hosted exterior bases with $\kappa t_I \ge m(c - 2\sigma)$. Hence its integer saving $z = 2t_I - m$ satisfies

$$\kappa z \ge dm, \qquad z \ge \frac{\rho m}{2000n} = \theta_s m. \tag{E.7}$$

The absorption and hosting phases still use disjoint edges. In their count, retain $e(S,H) = cb - a$ rather than replace it by cb. Equation (E.2a) therefore strengthens to

$$|Q| + e(G[S]) + 2m_W + a + \theta_s m \le cb + d_W + 2t_W(L + t_W). \tag{E.8}$$

Define the rational terminal comparator

$$F = cb - \binom{c}{2} - \binom{s+1}{2} + sc + \frac{t_W(c + b + s)}{3}, \qquad D_0 = Q_s(n) - F.$$

The two inequalities in (E.2e) give $D_0 \ge 0$, including their integer rounding. Substituting the definition of R_* into (E.8) gives $|Q| + a + \theta_s m + R_* \le F$. The lower bound on every order-four partition now yields

$$D_0 + a + \theta_s m + R_* \le \delta. \tag{E.9}$$

This proves the first account in (6.14), once the nonnegativity of R_* is retained. The partition here is the physical partition just constructed, not a separate optimizer.

Exceptions pay for their incidence costs. The proof of (E.2b) has enough room to replace every g_w by $\widetilde{g}_w = g_w + \rho/800$. We record why all three cases survive. Outside X, the degree and normalization estimates give

$$g_w \le -\rho/400 + \frac{10t_W}{3} + 2L.$$

Using $t_W \le 4\rho/10^{27}$, $L \le \rho/10^{13}+1$, and $\rho \ge 10^{50}$, the right-hand side is at most $-\rho/800$. Thus $\widetilde{g}_w \le 0$ there.

On X, replace e_1 by $\widetilde{e}_1 = e_1 + \rho/800$. The loss of more than $\rho/40$ light neighbors still gives $\widetilde{g}_w \le c - \rho/80$: the added $\rho/800$, the normalization error in $|\mathcal{L}| - n/3 - c$, and the original e_1 together use less than $\rho/80$. This proves the case $1 \le j \le s-\nu$ as before. In the case $j = 0$, all shifted contributions are nonpositive. In the remaining case, (E.2d) gives

$$\sum_{w\in W} \widetilde{g}_w \le (s-\nu)c - \frac{2n}{9} + \frac{4(s+1)\lambda}{10^{31}} + (s+1)\widetilde{e}_1.$$

Here $\widetilde{e}_1 \le 20\rho/10^{27} + 2\rho/10^{13} + 2 + \rho/800$. Since $n = \rho(s+1)^2$, $(s+1)\lambda \le n$, and $\binom{s+1}{2} \le \rho/(2\cdot 10^{50})$, the negative term $2n/9$ pays these errors and the binomial term. The same missing-pair bound used in E.1 therefore gives $\sum \widetilde{g}_w \le E$. Equivalently,

$$R_* \ge \frac{t_W \rho}{800} \ge 0, \qquad nt_W \le 800(s+1)^2 R_*. \tag{E.10}$$

This proves the strict budget from the constructor's estimates.

The remaining core excess is also paid. The normalization bounds imply $2(L+t_W) \le (c+b+s)/3$. Since the minimum subtracted in g_w is nonnegative, $g_w \le \deg(w) \le n$. Consequently,

$$E = R_* + \sum_{w\in W} g_w \le R_* + nt_W. \tag{E.11}$$

This account uses the bound on the same L supplied by the constructor.

An actual clique, with a paid displacement. First justify the missing-pair restriction on S. Suppose $s+1$ disjoint nonedge pairs have endpoints $B \subseteq S$, and let $U \subseteq \mathcal{L}$ be their common light neighbors. The normalized missing-link estimates and $|B| = 2(s+1)$ give

$$|U| \ge \rho/10^4 + s + 2.$$

Every clique in $\mathcal{L}$ has at most $\rho/10^4 + 1$ vertices, by the light exterior-degree bound. In the induced graph on $B \cup U$, each vertex of B therefore has more than s neighbors outside any clique of its neighborhood. Each vertex of U has the same property: its neighborhood contains B, and a clique can contain at most one endpoint of each missing pair. No vertex of this induced graph satisfies the rooted-defect condition with empty root, a contradiction. The common-neighbor estimates are essential to this contradiction.

For completeness, the finite core alternative used next is as follows. If $c \ge 20(s+1)^2$ and the complement of $G[S]$ has no matching of size $s+1$, then either S contains a clique of size $c-s$, or

$$c - s \le 2\left(e(G[S]) - \binom{c-s}{2}\right) = 2E. \tag{E.12}$$

One proves this by induction on s. For $s = 0$, S is a clique. If some vertex has at least $2s+1$ nonneighbors in S, deleting it leaves no missing matching of size s: such a matching would occupy only $2s$ vertices, leaving a nonneighbor to extend it. Apply induction with $s-1$; the target core size remains $c-s$, and any lower bound on the edges of the smaller set also holds in S. Otherwise the missing-edge graph has maximum degree at most $2s$. Repeatedly deleting the two endpoints of a missing edge removes at most $4s+1$ missing edges per step and stops after at most s steps. Thus it has at most $(4s+1)s$ edges. Substitution in $E = e(G[S]) - \binom{c-s}{2}$, using $c \ge 20(s+1)^2$, gives (E.12).

In the first case, choose the clique C of size $c-s$ and put $r = 0$. In the second case, the endpoints of a maximum missing matching meet every missing edge. Their complement is a

clique of size at least $c - 2s$; trim it to that size and put $r = s$. Then $|C| + s + r = c$. Since $c \geq 2s$, (E.12) gives $c \leq 4E$, and hence $rc \leq 4sE$. Normalization also gives $n \leq 4c$, so in both cases

$$rn \leq 16sE. \tag{E.13}$$

The construction preserves C as a clique of the original graph. The normalized size bounds also give $2 \leq |C|$ and $2|C| + s \leq n$, as required for the final root.

Counting the edits on the original vertex set. Start with the provisional template that has clique C, independent set $S \setminus C$, all links from S to H, and isolated vertices W. Because C is already a clique, comparison with G costs at most

$$a + m + \left(e(G[S]) - \binom{|C|}{2} \right) + nt_W.$$

Choose $D \subseteq S \setminus C$ with $|D| = s$, put $M = S \setminus (C \cup D)$, and move $M \cup W$ to the host class $J = H \cup M \cup W$. Here $|M| = r$. Every changed pair in this second comparison meets $M \cup W$, so it costs at most $n(r + t_W)$. Finally,

$$e(G[S]) - \binom{|C|}{2} = E + \binom{|C| + r}{2} - \binom{|C|}{2} \leq E + rn.$$

Combining the two comparisons gives

$$d_E(G, T(C, D, J)) \leq a + m + E + 2rn + 2nt_W. \tag{E.14}$$

One rn pays the reduced clique size in the interior count; the other pays the change of host class. Likewise, W is first isolated and then placed in the host class. These are upper bounds on two successive edits, not a claim that all charged pairs are distinct.

Combining the accounts. Equations (E.11), (E.13) and (E.14) imply

$$\begin{aligned} d_E(G, T) &\leq a + m + (1 + 32s)E + 2nt_W \\ &\leq a + m + (1 + 32s)R_* + (3 + 32s)nt_W \\ &\leq a + m + \big(1 + 32s + 800(3 + 32s)(s + 1)^2\big)R_*. \end{aligned}$$

The final coefficient is at most $38000(s + 1)^3$. Equation (E.9), with $D_0 \geq 0$, pays $a + m$ with at most $2000(s + 1)^2\delta$, and pays R_* with at most δ. Therefore

$$d_E(G, T) \leq 40000(s + 1)^3\delta.$$

This is the terminal estimate used in §6.6. Its constant pays for an actual clique of G; a cheaper comparison that merely repairs the core into a clique would not supply the same-root conclusion of Theorem C'.

Appendix F. Parameters and proof interfaces for the extensions

F.1. Explicit fixed-defect parameters

The explicit terminal and localization thresholds used below are those constructed in Appendix E.2. Write $\eta_s = \mathrm{etaNF}(s) > 0$, $L_s = \mathrm{NLoc}(\mathrm{NeditE}, s, \epsilon_s)$, and $F_s = \mathrm{Fexp}(\mathrm{NeditE}, s)$. Here NeditE is the chordal-edit threshold from the removal argument following de Joannis de Verclos [22], with the adaptations described in F.5. The function N_{far} is the explicit rounding threshold in Appendix C.3. Each function is fixed before the graph is chosen.

With ϵ_s, A_s as in (6.11), the actual stability parameters are

$$\begin{aligned}\gamma_s &= \tfrac{1}{2}\min\{1, \eta_s/4, \epsilon_s/4\},\\ N_s^{\deg} &= \lceil 10^{50}(s+1)^8\rceil + L_s + N_{\mathrm{far}}(\eta_s) + s + 5,\\ N_{0,s} &= N_s^{\deg} + F_s + s + 3 + \lceil 1/\epsilon_s\rceil + \lceil A_s\rceil + 1,\\ N_s^{\mathrm{stab}} &= 2N_{0,s} + F_s + s + 4.\end{aligned} \tag{F.1}$$

These are definitions of named, proved threshold functions, not numerically useful bounds. The local term $10^{50}(s+1)^8$ is only one summand; it must not be reported as the global threshold. The proof combines the removal, localization, far-rounding and retained-terminal results. Thus (F.1) contains no assumed realizer or localization theorem.

For clarity, the deletion invariant used in §6.6 is

$$0 \le \delta \le \gamma_0 t(t - N_{0,s}), \qquad \gamma_0 = \min\{1, \eta_s/4, \epsilon_s/4\}.$$

When a low-degree vertex is removed, (6.15) decreases the deficit by at least $\epsilon_s t - 1/3$. The change of the right-hand side is $\gamma_0(2t - 1 - N_{0,s}) \le \epsilon_s t/2$. Since $\epsilon_s N_{0,s} \ge 1$, this pays the change and leaves a nonnegative child deficit. If deletion were to reach the bottom order, the invariant would force a zero deficit while the same deletion estimate forces a negative child deficit, a contradiction. This is why a finite small-order case does not introduce an additive error in stability.

F.2. Fractional star accounting and small orders

For $n > 0$, take an optimal mixed dual y and use signed prices $z_e = 1 - y_e$. The triangle and K_4 constraints give signed clique sums at most one. The maximum positive clique-star argument of [15, Lemma 3.1], followed by rooted elimination, yields an integer $c < n$ and a price α with $0 \le \alpha \le c$ such that

$$\begin{aligned}F_4(G) &\le (n - 2c + 1)\alpha + \binom{c}{2} + \sum_{j=0}^{n-c-1} \min(s, j),\\ F_4(G) &\le (n-c)\alpha + \frac{1}{6}\binom{c}{2} + s(n-c) \qquad\qquad (c \ge 4).\end{aligned} \tag{F.2}$$

The certified optimum is used here: a feasible dual value alone is not identified with $W^*(G)$.

If $n - c \ge s$, the first exception sum is exactly

$$s(n-c) - \binom{s+1}{2}. \tag{F.3}$$

On the low-star branch $2c \le n+1$, the coefficient of α is nonnegative, so $\alpha \le c$ can be substituted in the first row. In the band $4s \le n$, (F.3) then gives

$$F_4(G) \le B_{n-s}(c) + ns - \binom{s+1}{2} \le M(n-s) + ns - \binom{s+1}{2}.$$

This uses an integer baseline inequality before the final comparison, not a floor applied to a fractional value. On the high-star branch, compare α with $5c/12$; the second row is used below that value and the first above it. Both give the envelope

$$F_4(G) \le \frac{c(5n - 4c - 1)}{12} + s(n-c). \tag{F.4}$$

The rational comparisons establish the two bands in Proposition 6.4. For $n < 8$, the remaining integer parameter ranges are checked against this same envelope. The empty graph and the cases not requiring a K_4 constraint use the first row or the edge-count bound. No enumeration of small graphs, floating-point optimum, or external solver certificate is needed for these orders.

F.3. Uniform localization through a chordal edit

The structural part of Theorem 6.5 does not use only (F.2). It also needs to turn small rooted defect into a small chordal edit. For each accuracy $\xi > 0$, the removal construction gives an explicit sample size $m = m(\xi)$. Set

$$\theta(\xi) = \frac{1}{16m^{m+1} + 1}, \qquad N(\xi) = \max\{m, 1\}. \tag{F.5}$$

For $n \geq N(\xi)$ and $s \leq \theta(\xi)n$, a graph of rooted defect at most s is within ξn^2 edge edits of a chordal graph. The underlying removal proof follows [22], with the adaptations specified in F.5.

The next lemma recovers a clique in the original graph. For a vertex set D, let $\mathrm{ordNE}(G, D)$ count ordered pairs of distinct, nonadjacent vertices of D, and write $x_+ = \max\{x, 0\}$.

Lemma F.3a (clique recovery with additive defect). If $\mathrm{rsd}(G) \leq s$, every $D \subseteq V(G)$ contains a clique C of G such that

$$(|D| - |C| - s)_+^2 \leq \mathrm{ordNE}(G, D). \tag{F.5a}$$

Proof. Induct on $|D|$; the empty set is immediate. Apply the rooted condition to D, with empty prescribed root. There is a vertex $v \in D$ whose neighbors in D contain a clique P with at most s neighbors outside P. Let y count the nonneighbors of v in $D \setminus \{v\}$. Removing v removes precisely the two orientations of those y nonedges, so

$$\mathrm{ordNE}(G, D) = \mathrm{ordNE}(G, D \setminus \{v\}) + 2y.$$

By induction, $D \setminus \{v\}$ contains a clique C' with $x^2 \leq \mathrm{ordNE}(G, D \setminus \{v\})$, where $x = (|D| - 1 - |C'| - s)_+$. If $y \geq x + 1$, take $C = C'$. Then

$$(|D| - |C'| - s)_+^2 \leq (x + 1)^2 \leq x^2 + 2y \leq \mathrm{ordNE}(G, D).$$

Otherwise $y \leq x$, since both are integers. Take $C = P \cup \{v\}$, which is a clique. The degree decomposition gives $|P| + s \geq |D| - 1 - y$, so $(|D| - |C| - s)_+ \leq y \leq x$. Squaring and using the induction bound proves (F.5a) in this case as well.

Corollary F.3b (recovery after edge deletion). Suppose $\mathrm{rsd}(G) \leq s$, G, H have the same vertex set, and A is a clique of H. If $u \geq 0$ and

$$2|E(H) \setminus E(G)| \leq u^2,$$

then A contains a clique R of G with $|A \setminus R| \leq s + u$.

Proof. Every unordered nonedge of G within A is an edge of H missing from G. Consequently, $\mathrm{ordNE}(G, A) \leq 2|E(H) \setminus E(G)|$. Apply (F.5a); taking nonnegative square roots gives $(|A| - |R| - s)_+ \leq u$, which is the required estimate.

The displayed proof applies to real $u \geq 0$. When $s = 0$, (F.5a) reduces to the chordal clique-extraction estimate (F.9). The loss in s is additive, rather than a factor multiplying the square-root term.

Now let H be the chordal graph obtained above and put $t = d_E(G, H)$. Repairing bounded-order partitions transfers the integral lower bound with deficit at most $\delta + 4t$. Apply Theorem 6.1 in H. Its clique need not be a clique of G; Corollary F.3b removes at most $s + u$ vertices for any $u \geq 0$ with $2t \leq u^2$, producing a clique R in G itself. Counting changed edges and incidences yields

$$\begin{aligned} e(G - R) + A_R &\leq 16\delta + 65t + n(s + u), \\ M(n) - B_n(|R|) &\leq \delta + 4t + 2n(s + u). \end{aligned} \tag{F.6}$$

Here δ is measured relative to $M(n)$. To start from the hypothesis at $n^2/6$ in Theorem 6.5, increasing its deficit by $n/6$ suffices and preserves an $o(n^2)$ error. Choose ξ, the defect ratio and the near-extremality margin sufficiently small before n and G; then u can be chosen as a small multiple of n. Formula (F.6) proves uniform edit and baseline control. The comparator parabola turns the latter into a size window, at a squared accuracy scale. Minimizing (6.27) then gives the one-root sequence statement. This order of choices prevents both a hidden fixed-s threshold and a root depending on the requested partition.

F.4. Sharpness of the local coefficient

The factor ten in the local inequality $\binom{a+b}{2} \le 10d(a,b)$, for positive defect, is sharp: the profile $(a,b) = (3,2)$ has ten edges and defect one. This does not prove optimality of 480, A_s, or the global thresholds.

Appendix E.4 derives the retained terminal accounts used in §6.6, F.3 proves clique recovery, and F.5 explains the three-case assembly of the adapted pinned-sample bound.

F.5. Auxiliary lemmas and adaptations of chordal removal

The removal implementation follows [22], but is not a literal translation of its statements and constants. Four finite lemmas make the changes relevant to the present application explicit.

Induced-cycle count. If $\mathrm{rsd}(G) \le s$, then for every $k \ge 4$,

$$\mathrm{indCopies}(C_k, G) \le 2ks\, n^{k-1}. \tag{F.7}$$

Here indCopies counts labelled induced embeddings of the k-cycle, not unlabelled cycle vertex sets. This convention matters in using (F.7) in the sample union bound. It supplies the cycle-count input directly from rooted defect.

Co-bipartite edit bound. If the vertex set is the union of two cliques and $\mathrm{rsd}(G) \le s$, then there is a chordal H on the same vertices with

$$d_E(G,H) \le 2\sqrt{s}\, n\sqrt{n} + n. \tag{F.8}$$

The proof orders one clique, bounds the inversions in the other clique's adjacency rows, and replaces each row by a threshold row. The resulting nested cross-neighborhoods give a chordal graph. The estimate concerns edit distance, not a clique partition of the original graph.

Clique extraction by elimination. If $G[D]$ is chordal, $d = |D|$, and M is the number of unordered nonedges in D, then D contains a clique C satisfying

$$(d - |C|)^2 \le 2M, \qquad |C| \ge d - \sqrt{2M}. \tag{F.9}$$

The proof is an induction by simplicial elimination. Counting ordered nonedges instead gives twice the unordered count, which explains the factor two. This replaces the use of [22, Lemma 4] inside the near-simplicial repair; it is not presented as a new version of the full theorem cited there.

Near-simplicial repair. Partition $V(G) = X \sqcup Y$, assume $G[Y]$ is chordal, and let $p_G(x)$ count unordered nonedges within $N_G(x)$. If $\varepsilon \ge 0$ and $p_G(x) \le \varepsilon n^2$ for every $x \in X$, then there is a chordal H with

$$d_E(G,H) \le 8\sqrt{\varepsilon}\, n^2, \qquad H[Y] = G[Y]. \tag{F.10}$$

Compared with [22, Lemma 3], the constant is eight rather than six, while the condition $n \ge 1/\varepsilon$ is not required. Neither version is described as uniformly stronger: the hypotheses and constants differ.

The rest of the implementation changes the representation and the numerical schedule as follows.

Step of [22]	Implementation used here
Sampling	Samples are sequences with replacement. Passage is made from sequence counts to vertex subsets for the upward-closed property at issue.
Pinned representations	The definition allows a path-preserving map from the prescribed tree into another rooted tree. The counting proof handles this enlarged class of representations; its sample bound also changes.
Tree model	Finite parent maps and node sets replace topological trees. The parameter K counts nodes, not leaves. Sections are indexed by nodes, so the corresponding Claim 1 is not imported.
Set-coloring count	Affected vertices are counted explicitly. Small and degenerate parameter cases are separated instead of importing the published logarithmic bounds verbatim.
Pinned sample bound	The chosen size is $m_{11}(\varepsilon, K) = \lceil 2^{58}(K+1)^{15}/\varepsilon^{12} \rceil$.
Final union bound	Use $q = \lceil 102400/\varepsilon^2 \rceil^2$ sampled pairs and $\lceil \log_2(4\, n_{\mathrm{codes}}) \rceil$ independent blocks, with n_{codes} the explicitly counted family of gate data.

Table 7. Adaptations of [22] relevant to the explicit threshold. The larger parameter bounds do not improve its sample-size exponent. The proof remains an implementation of its removal strategy, with these finite replacements and parameter choices.

Assembly of the adapted pinned-sample bound. We give the three cases of the assembly, since changing the representation and the sampling convention requires checking the counting argument, not merely citing the published lemma. Fix a rooted tree Γ with K nodes and a prescribed pin $x_v \in V(\Gamma)$ for every vertex of G. A sample is *pinned* if its induced graph admits a representation by subtrees of another rooted tree T, together with a path-preserving map $\iota : \Gamma \to T$, such that the subtree for v contains $\iota(x_v)$. This is the enlarged notion in Table 7.

Assume G is ε-far from chordal: every chordal graph on its vertex set has edit distance greater than εn^2. Comparison with the empty graph forces $n \geq 1$ and $\varepsilon < 1/2$; the pin map then forces $K \geq 1$. For $\varepsilon > 0$, choose

$$y = \left\lceil \frac{256K^2}{\varepsilon^2} \right\rceil, \qquad \delta_{11} = \frac{1}{4y^2}, \qquad B = \left\lfloor \frac{8K}{\delta_{11}} \right\rfloor, \qquad \eta = \frac{\varepsilon}{2K},$$
$$m = m_{11}(\varepsilon, K) = \left\lceil \frac{2^{58}(K+1)^{15}}{\varepsilon^{12}} \right\rceil. \tag{F.11}$$

We must show that at most half of the n^m ordered samples of length m, with replacement, are pinned.

An admissible set-coloring assigns to v a subtree $\psi(v)$ of Γ containing x_v. For an edge uv, compatibility requires the path from x_u to x_v to lie in $\psi(u) \cup \psi(v)$; for a nonedge it requires $\psi(u) \cap \psi(v) = \varnothing$. Let $\mathrm{conf}(\psi)$ count ordered incompatible pairs of distinct vertices. A pinned sample induces a compatible set-coloring on its vertex set.

Case A: every admissible coloring has at least $\delta_{11} n^2$ conflicts. Apply the adapted set-coloring counting theorem with sample size m and B rounds. The numerical conditions are $8K/\delta_{11} < B + 1$ and, for the parameters (F.11),

$$2(1 - \delta_{11}/4)^m \left(1 + \frac{m 2^K}{1 - \delta_{11}/4}\right)^B \leq 1.$$

It follows that at most half of all samples admit a compatible set-coloring. Pinned samples form a subset, so the same bound holds for them.

Case B1: some admissible ψ has fewer than $\delta_{11}n^2$ conflicts, but one section is far from every chain model. Sections are indexed by the K nodes of Γ; each section has two vertex classes and a rank model for their cross-adjacencies. In this case one section has more than ηn^2 mismatches for every rank assignment. The section-counting lemma bounds the number of length-y samples inducing a chordal graph by

$$(y+1)(1-\eta)^{y-1}n^y + y^2\,\mathrm{conf}(\psi)n^{y-2}.$$

The parameter estimate gives $(y+1)(1-\eta)^{y-1} \le 1/4$. The second summand is at most $n^y/4$, since $y^2\delta_{11} = 1/4$. Thus at most half of the length-y samples are chordal. We have $y \le m$; every pinned length-m sample induces a chordal graph, and its prefix of length y does too. Each prefix has n^{m-y} extensions. Hence at most $n^m/2$ full samples are pinned. This argument uses sequences throughout and does not assume their entries are distinct.

Case B2: every section has a rank assignment with at most ηn^2 mismatches. Normalize each rank by replacing it with the number of vertices of strictly smaller rank. This preserves all strict rank comparisons, and therefore all section mismatches, while bounding every rank by n. Subdivide the prescribed tree using this common bound and glue the section models. The resulting graph F is an intersection graph of subtrees, hence chordal. Write mis_c for the number of mismatches in section c. The gluing estimate charges every changed edge either to a conflict or to one of these mismatches:

$$\begin{aligned} d_E(G,F) &\le \mathrm{conf}(\psi) + \sum_{c\in V(\Gamma)} \mathrm{mis}_c \\ &< \delta_{11}n^2 + K\eta n^2 \le \varepsilon n^2. \end{aligned} \tag{F.12}$$

Here $\delta_{11} \le \varepsilon/2$, by (F.11). This contradicts the assumed distance from chordality, so Case B2 cannot occur.

The alternatives are exhaustive: either every coloring has many conflicts, or one has few, and its sections either include a distant one or all have close rank models. The set-coloring theorem, the section-counting lemma and the subtree-gluing lemma retain their roles from the removal strategy of [22]; the argument above identifies how their adapted versions fit together. It does not assert that the original published Lemma 11 has the modified statement or constants.

Appendix G. Packing gaps and limits of cleanup

G.1. The all-order bound and packing gaps

Does $c_4(G) \le M(n)$ hold for every chordal graph of every order, that is, can one take $b = 0$ in Theorem A? The threshold in this proof limits the proof; it is not evidence of an exception to the statement. For the complete family there is a construction without a threshold: $c_4(K_n) \le M(n)$ for every $n \ge 0$, as proved in Appendix D. Unlike $\mathrm{cp}(K_n) = 1$ for $n \ge 2$, this assertion requires control of piece size.

The proof neither requires nor establishes a universal linear bound for the mixed gap $W^*(G) - W(G)$, where $W(G)$ is the optimal integral mixed gain. It compares the loss of the construction with the margin available in each instance. The general growth rate of this gap is studied separately; it is not a hypothesis of this paper.

There is a complementary result for the **triangle gap** with bounded clique number. Let $d \ge 0$ be an integer and G a chordal graph with no clique of order $d+2$. Every fractional triangle

packing x satisfies

$$\sum_T x_T \le \nu_3(G) + \left(10 + \frac{d}{2}\right) n. \tag{G.1}$$

Here $\nu_3(G)$ counts edge-disjoint triangles; the fractional objective also counts triangles, without the gain factor two. Taking the optimum gives $\nu_3^*(G) - \nu_3(G) \le (10 + d/2)n$. For fixed d, the loss is linear; this does not give a uniform constant when the clique number grows.

The proof uses a subtree representation and separates the mass of local pieces from mass crossing the joins. Each triangle of the latter kind uses two joining edges; summing their capacities pays for its mass with half the count of those edges. The bound d controls that count and contributes the term $dn/2$ in addition to the local loss $10n$.

The hypothesis can equivalently be expressed through a clique tree whose bags have at most $d+1$ vertices: for chordal graphs this is equivalent to excluding K_{d+2}. Result (G.1) does not bound the **mixed** gap for K_3/K_4 or replace the rounding in Section 3.

G.2. A quantitative obstruction to codegree cleanup

The threshold in Theorem 3.1 comes from the regularity construction in Appendix C. A complementary argument, separate from that proof, bounds what can be achieved through another cleanup interface. The two objects must be distinguished: proving that a cleanup suffices for rounding does not prove that every rounding method must perform it.

Fix $\varepsilon > 0$, $\gamma \le 1/2$ and a constant C_0. Consider the following contract. For every graph G of order $n \ge N_0$ and every fractional mixed packing x whose triangular mass is at least $\varepsilon n^2/30 - 1$, it requires another packing y on the same graph with triangular mass at least C_0, gain loss at most $\varepsilon n^2/4$ and weighted codegree at most γ. This codegree is the sum of the weights of copies simultaneously containing two distinct edges of the graph.

Proposition G.1 (limitation of this contract). If N_0 satisfies the preceding contract, $t \ge 4$ and $7\varepsilon \le e^{-t}$, then

$$N_0 > \exp(t^2/16). \tag{G.2}$$

In particular, if $\varepsilon \le 1/(7e^{34})$, then $N_0 > e^{72}$.

Proof. Call a graph rigid if each of its edges belongs to exactly one triangle and it has no K_4 copies. If it has T triangles, assigning weight one to each gives gain $2T$. Two edges of the same triangle can receive joint weight only through that copy. Thus every y with codegree at most γ assigns it weight at most γ, and its gain is at most $2\gamma T$. The forced loss is at least $2(1-\gamma)T$. For an order satisfying the mass hypothesis, the contract imposes

$$2(1-\gamma)T \le \frac{\varepsilon}{4} n^2. \tag{G.3}$$

Thus a uniform cleanup of this kind quantitatively controls the density of rigid systems, a problem of $(6,3)$ type.

The tripartite construction of Ruzsa–Szemerédi [18], applied to a set $A \subseteq \{0, \dots, M-1\}$ with no three-term arithmetic progressions, gives a rigid graph of order $6M+3$ with at least $(2M+1)|A|$ triangles. The version of Behrend's bound [17] used here guarantees $|A| \ge M \exp(-4\sqrt{\log M})$. If $7\varepsilon \le \exp(-4\sqrt{\log M})$ and $M \ge 1$, this count satisfies the mass hypothesis and contradicts (G.3): the order $6M+3$ must be smaller than N_0.

Now take $M = \lfloor \exp(t^2/16) \rfloor$. Then $M \ge 1$, $\log M \le t^2/16$ and $4\sqrt{\log M} \le t$. The preceding paragraph applies. Since $\exp(t^2/16) < M + 1 \le 6M + 3$, (G.2) follows. For $t = 34$, $t^2/16 = 72.25 > 72$, giving the final assertion.

For $t = \log(1/(7\varepsilon))$, (G.2) grows faster than every fixed power of $1/\varepsilon$. This rules out a polynomial threshold for **this contract**, not for every proof of Erdős 81. It does not show that regularity is the only possible method, that a tower is necessary, or that e^{72} is a lower bound on the threshold in Theorem 3.1. The complement proves cleanup $\Rightarrow$ rounding; transferring its barrier to another rounding method would require the reverse implication or a quantitative adapter.

What changes under chordality. Both the cleanup and the rounding statements quantify over all graphs, whereas the far-branch assembly uses them on the original chordal graph. The rigid family above does not give the same obstruction in that class. Indeed, let G be chordal, with no K_4, and $n \geq 2$. In a perfect elimination ordering each vertex has at most two later neighbors; otherwise it and three such neighbors would form a K_4. Counting each edge at its earlier endpoint, the first $n-2$ vertices contribute at most two each, the penultimate at most one, and the last none. Hence

$$e(G) \leq 2(n-2) + 1 = 2n - 3. \tag{G.4}$$

If G is also rigid, its T triangles are edge-disjoint. Thus $3T \leq e(G)$, and any feasible fractional triangular mass is at most T. The mass hypothesis of the contract would require

$$\frac{\varepsilon n^2}{30} - 1 \leq T \leq \frac{2n-3}{3} \quad \Longrightarrow \quad n \leq \frac{20}{\varepsilon}. \tag{G.5}$$

Cancellation of the constant terms explains the final threshold. Above it, no rigid chordal graph satisfies the mass hypothesis. In particular, the dense instances used to obtain the superpolynomial barrier cannot be chordal. The same is not asserted of every small or degenerate instance of the tripartite construction.

This count delimits the obstruction; it proves neither a chordal cleanup nor an upper bound $O(1/\varepsilon)$ on its threshold. Studying that weaker contract is a question distinct both from the universal barrier and from the bounded-clique triangle gap of Appendix G.1.

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Use of artificial intelligence and computational tools

Claude, by Anthropic, and ChatGPT/Codex, by OpenAI, were used to explore and check arguments and to prepare the manuscript. Aristotle, by Harmonic, contributed to exploring approaches, searching for counterexamples, developing proofs, and formalizing and reviewing them in Lean; its role was not limited to translating finished proofs.

Certo [13], https://github.com/jtraverso/certo-math, and Jacobian [14], https://github.com/morluto/jacobian, were the computational tools used to explore instances, check identities, and assess candidates. Jacobian is also credited by the project behind [5]; its use here is acknowledged separately from the logical dependencies of our formal proof. Computations were reviewed within their stated scope; a finite search does not replace a universal proof.

The author remains responsible for the arguments, citations, code, and presentation. No AI system is listed as an author.

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